

ADVANCED NETWORK SECURITY AND ARCHITECTURES

METI

Report Module 2 [EN]

Authors:

José Oliveira (103771)

Afonso Mendes (116024)

Tiago Videira (116069)

jose.novais.oliveira@tecnico.ulisboa.pt

afonsofmedes@tecnico.ulisboa.pt

tiago.g.videira@tecnico.ulisboa.pt

Group 3

2025/2026 – 2nd Semester, P4

Contents

1	Introduction	3
2	IKEv1 with Pre-Shared Keys	4
2.1	Introduction	4
2.2	AH Tunnel Operation	4
2.3	OSPF Traffic Analysis	4
2.4	IKEv1 and ISAKMP Negotiation	6
2.4.1	ISAKMP Informational Messages	6
2.4.2	IKE Phase 1 – Main Mode	7
2.4.3	IKE Phase 2 – Quick Mode	8
2.5	Negotiation Failure Analysis	9
2.6	ESP Tunnel Operation	10
2.7	Firewall Requirements for IPSec	11
3	IKEv1 with Digital Signatures	12
3.1	Introduction	12
3.2	Topology Overview	12
3.3	PKI Installation Verification	12
3.3.1	CA Configuration and Certificate (RA)	12
3.3.2	Router Certificates (R1)	13
3.3.3	RSA Public Keys (R1 and R2)	14
3.4	SCEP Enrollment Analysis	16
3.4.1	CA Certificate Retrieval	16
3.4.2	Certificate Enrollment Request	16
3.5	Encrypted ICMP Traffic Analysis	18
3.6	ISAKMP and SA Re-establishment Analysis	20
3.7	Peer Public Key Storage Verification	21
3.8	SCEP Messages Observed After Router Restart	22
3.9	Certificate Revocation and Authentication Failure	23
3.9.1	SCEP and Certificate Validation Traffic	23
3.9.2	Failed ISAKMP Negotiation	24
3.9.3	Absence of IPSec Security Associations	25
3.9.4	Explanation of Communication Failure	26
4	IPSec with IKEv2	26
4.1	Introduction	26
4.2	IKEv2 and IPSec Configuration Verification	26
4.3	Tunnel Establishment and Connectivity Test	28
4.4	Analysis of IKE_SA_INIT Exchange	29
4.5	SA Deletion and Re-establishment	32
4.6	IKEv2 Security Association Rekeying	32
4.7	IPSec Security Association Rekeying	35

5	DMVPN Phase 3	37
5.1	Topology	37
5.2	DMVPN Phase 3 Configuration	37
5.3	Role of <code>ip summary-address rip</code>	37
5.4	NHRP Redirects and Shortcuts	38
5.5	Traffic Analysis	39
5.6	Wireshark Analysis	39
5.7	Role of RIP Split Horizon	39
5.8	Comparison Between DMVPN Phases	39
5.9	Experimental Validation	40
5.10	Comparison Between AI Explanation and Experimental Results	40
6	VRFs and BGP-MPLS L3VPNs	40
6.1	Introduction	40
6.2	Ping Between Sites and Loopback Sourcing	41
6.3	PE1 Routing Tables	41
6.4	MPLS Labels During Ping Analysis	43
6.5	BGP Session Establishment and VPNv4 Route Advertisement	46

1 Introduction

2 IKEv1 with Pre-Shared Keys

2.1 Introduction

In this experiment, IPSec tunnels using IKEv1 with Pre-Shared Keys (PSK) were analyzed. Both AH and ESP protocols were tested in order to observe the establishment of IPSec Security Associations (SAs), OSPF traffic protection, and the exchanged ISAKMP messages during tunnel negotiation.

2.2 AH Tunnel Operation

Initially, the IPSec tunnel was configured using the Authentication Header (AH) protocol. ICMP traffic exchanged between PC1 and PC2 was captured using Wireshark.

Figure 1 shows an AH-protected ICMP packet. The AH header includes fields such as the Security Parameters Index (SPI), Sequence Number, and Integrity Check Value (ICV). The SPI identifies the Security Association being used, the sequence number prevents replay attacks, and the ICV guarantees packet integrity and authentication.

Unlike ESP, AH does not encrypt the payload. Therefore, the original ICMP packet remains visible after the AH header.

No.	Time	Source	Destination	Protocol	Leng	Info
67	16:48:28.535311	192.168.2.1	224.0.0.5	OSPF	138	Hello Packet
69	16:48:30.699711	192.168.3.2	224.0.0.5	OSPF	138	Hello Packet
73	16:48:33.199506	192.168.1.100	192.168.4.100	ICMP	158	Echo (ping) request id=0x0000, seq=0/0, ttl=253 (no response found!)
74	16:48:34.170234	192.168.1.100	192.168.4.100	ICMP	158	Echo (ping) request id=0x0000, seq=1/256, ttl=253 (no response found!)
75	16:48:36.170697	192.168.1.100	192.168.4.100	ICMP	158	Echo (ping) request id=0x0000, seq=2/512, ttl=253 (reply in 76)
76	16:48:36.227667	192.168.4.100	192.168.1.100	ICMP	158	Echo (ping) reply id=0x0000, seq=2/512, ttl=253 (request in 75)
77	16:48:36.262752	192.168.1.100	192.168.4.100	ICMP	158	Echo (ping) request id=0x0000, seq=3/768, ttl=253 (reply in 78)
78	16:48:36.331440	192.168.4.100	192.168.1.100	ICMP	158	Echo (ping) reply id=0x0000, seq=3/768, ttl=253 (request in 77)
79	16:48:36.375843	192.168.1.100	192.168.4.100	ICMP	158	Echo (ping) request id=0x0000, seq=4/1024, ttl=253 (reply in 80)
80	16:48:36.445216	192.168.1.100	192.168.4.100	ICMP	158	Echo (ping) reply id=0x0000, seq=4/1024, ttl=253 (request in 79)
81	16:48:38.090641	192.168.2.1	224.0.0.5	OSPF	138	Hello Packet
82	16:48:38.798171	192.168.1.100	192.168.4.100	ICMP	158	Echo (ping) request id=0x0001, seq=0/0, ttl=253 (reply in 83)
83	16:48:38.843275	192.168.4.100	192.168.1.100	ICMP	158	Echo (ping) reply id=0x0001, seq=0/0, ttl=253 (request in 82)
Frame 77: Packet, 158 bytes on wire (1264 bits), 158 bytes captured (1264 bits) on interface -, id 0						
Ethernet II, Src: 0c:d6:9b:f8:00:00 (0c:d6:9b:f8:00:00), Dst: ca:01:08:63:00:00 (ca:01:08:63:00:00)						
Internet Protocol Version 4, Src: 1.1.1.1, Dst: 2.2.2.2						
Authentication Header						
Next header: IPIP (4)						
Length: 4 (24 bytes)						
Reserved: 0000						
AH SPI: 0xc132efa1						
AH Sequence: 50						
AH ICV: 8479a3b6348f138b72cbf5e6						
Internet Protocol Version 4, Src: 192.168.1.100, Dst: 192.168.4.100						
Internet Control Message Protocol						

Figure 1: AH protecting ICMP traffic between PC1 and PC2

2.3 OSPF Traffic Analysis

OSPF traffic exchanged through the tunnel was also analyzed.

Figure 2 shows a normal OSPF Hello packet before IPSec protection. The packet includes the Router ID, Area ID, Hello Interval, Dead Interval, and neighbor information required for adjacency establishment.

161	16:50:27.086521	200.1.1.10	224.0.0.5	OSPF	94	Hello Packet
162	16:50:27.346867	200.1.1.1	224.0.0.5	OSPF	94	Hello Packet
163	16:50:28.006268	192.168.3.2	224.0.0.5	OSPF	138	Hello Packet
166	16:50:31.079919	192.168.2.1	224.0.0.5	OSPF	138	Hello Packet
167	16:50:36.734564	200.1.1.10	224.0.0.5	OSPF	94	Hello Packet

>	Frame 161: Packet, 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface -, id 0
>	Ethernet II, Src: ca:01:08:63:00:00 (ca:01:08:63:00:00), Dst: IPv4mcast_05 (01:00:5e:00:00:05)
>	Internet Protocol Version 4, Src: 200.1.1.10, Dst: 224.0.0.5
✓	Open Shortest Path First
✓	OSPF Header
	Version: 2
	Message Type: Hello Packet (1)
	Packet Length: 48
	Source OSPF Router: 200.2.2.10
	Area ID: 0.0.0.0 (Backbone)
	Checksum: 0xc77c [correct]
	Instance ID: Base IPv4 Unicast Instance (0)
	Auth Type: Null (0)
	Auth Data (none): 0000000000000000
✓	OSPF Hello Packet
	Network Mask: 255.255.255.0
	Hello Interval [sec]: 10
>	Options: 0x12, (L) LLS Data block, (E) External Routing
	Router Priority: 1
	Router Dead Interval [sec]: 40
	Designated Router: 200.1.1.10
	Backup Designated Router: 200.1.1.1
	Active Neighbor: 200.1.1.1
>	OSPF LLS Data Block

Figure 2: OSPF Hello packet without IPSec protection

Figure 3 shows the same OSPF traffic protected using AH. The Authentication Header appears before the OSPF packet, proving that the packet integrity and authentication are protected. However, the OSPF payload itself remains visible because AH does not provide confidentiality.

163	16:50:28.006268	192.168.3.2	224.0.0.5	OSPF	138	Hello Packet
166	16:50:31.079919	192.168.2.1	224.0.0.5	OSPF	138	Hello Packet
167	16:50:36.734564	200.1.1.10	224.0.0.5	OSPF	94	Hello Packet
168	16:50:37.564727	192.168.3.2	224.0.0.5	OSPF	138	Hello Packet

```

> Frame 163: Packet, 138 bytes on wire (1104 bits), 138 bytes captured (1104 bits) on interface -, id 0
> Ethernet II, Src: ca:01:08:63:00:00 (ca:01:08:63:00:00), Dst: 0c:d6:9b:f8:00:00 (0c:d6:9b:f8:00:00)
> Internet Protocol Version 4, Src: 2.2.2.2, Dst: 1.1.1.1
  Authentication Header
    Next header: IPIP (4)
    Length: 4 (24 bytes)
    Reserved: 0000
    AH SPI: 0xa9a6b9aa
    AH Sequence: 63
    AH ICV: 934e067a06754f65bac5a356
> Internet Protocol Version 4, Src: 192.168.3.2, Dst: 224.0.0.5
  Open Shortest Path First
    OSPF Header
      Version: 2
      Message Type: Hello Packet (1)
      Packet Length: 48
      Source OSPF Router: 2.2.2.2
      Area ID: 0.0.0.0 (Backbone)
      Checksum: 0xe595 [correct]
      Instance ID: Base IPv4 Unicast Instance (0)
      Auth Type: Null (0)
      Auth Data (none): 0000000000000000
    OSPF Hello Packet
      Network Mask: 0.0.0.0
      Hello Interval [sec]: 10
      Options: 0x12, (L) LLS Data block, (E) External Routing
      Router Priority: 1
      Router Dead Interval [sec]: 40
      Designated Router: 0.0.0.0
      Backup Designated Router: 0.0.0.0
      Active Neighbor: 1.1.1.1
    OSPF LLS Data Block

```

Figure 3: OSPF Hello packet protected with AH

2.4 IKEv1 and ISAKMP Negotiation

The IPsec Security Associations were cleared using the command:

```
clear crypto session
```

This operation forced the routers to delete the existing IPsec tunnel and establish a new one. During this process, Wireshark captured the exchanged ISAKMP packets.

2.4.1 ISAKMP Informational Messages

Before re-establishing the tunnel, ISAKMP Informational messages were exchanged between peers. These messages are used to maintain or delete Security Associations.

Figure 4 shows an Informational exchange containing encrypted data.

41	17:24:43.971851	1.1.1.1	2.2.2.2	ISAKMP	118	Informational
42	17:24:44.038276	1.1.1.1	2.2.2.2	ISAKMP	118	Informational
43	17:24:44.264407	1.1.1.1	2.2.2.2	ISAKMP	134	Informational
44	17:24:45.115767	2.2.2.2	1.1.1.1	ISAKMP	134	Informational
45	17:24:45.397728	1.1.1.1	2.2.2.2	ISAKMP	210	Identity Protection (Main Mode)

```

> Frame 41: Packet, 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface -, id 0
> Ethernet II, Src: 0c:d6:9b:f8:00:00 (0c:d6:9b:f8:00:00), Dst: ca:01:08:63:00:00 (ca:01:08:63:00:00)
> Internet Protocol Version 4, Src: 1.1.1.1, Dst: 2.2.2.2
> User Datagram Protocol, Src Port: 500, Dst Port: 500
> Internet Security Association and Key Management Protocol
  Initiator SPI: 77ece4687be4e5cf
  Responder SPI: c2f86656ee8a5d4c
  Next payload: Hash (8)
  > Version: 1.0
  Exchange type: Informational (5)
  > Flags: 0x01
  Message ID: 0x48606521
  Length: 76
  Encrypted Data (48 bytes)

```

Figure 4: ISAKMP Informational message

2.4.2 IKE Phase 1 – Main Mode

IKE Phase 1 establishes the ISAKMP Security Association used to securely negotiate IPsec parameters.

Figure 5 shows the initial Main Mode exchange. The Security Association payload contains the proposed cryptographic algorithms and authentication methods supported by the initiator.

45	17:24:45.397728	1.1.1.1	2.2.2.2	ISAKMP	210	Identity Protection (Main Mode)
46	17:24:45.753871	2.2.2.2	1.1.1.1	ISAKMP	150	Identity Protection (Main Mode)
47	17:24:45.833812	1.1.1.1	2.2.2.2	ISAKMP	390	Identity Protection (Main Mode)

```

> Frame 45: Packet, 210 bytes on wire (1680 bits), 210 bytes captured (1680 bits) on interface -, id 0
> Ethernet II, Src: 0c:d6:9b:f8:00:00 (0c:d6:9b:f8:00:00), Dst: ca:01:08:63:00:00 (ca:01:08:63:00:00)
> Internet Protocol Version 4, Src: 1.1.1.1, Dst: 2.2.2.2
> User Datagram Protocol, Src Port: 500, Dst Port: 500
> Internet Security Association and Key Management Protocol
  Initiator SPI: 77ece468cdcf914d
  Responder SPI: 0000000000000000
  Next payload: Security Association (1)
  > Version: 1.0
  Exchange type: Identity Protection (Main Mode) (2)
  > Flags: 0x00
  Message ID: 0x00000000
  Length: 168
  > Payload: Security Association (1)
  > Payload: Vendor ID (13) : RFC 3947 Negotiation of NAT-Traversal in the IKE
  > Payload: Vendor ID (13) : draft-ietf-ipsec-nat-t-ike-07
  > Payload: Vendor ID (13) : draft-ietf-ipsec-nat-t-ike-03
  > Payload: Vendor ID (13) : draft-ietf-ipsec-nat-t-ike-02\n

```

Figure 5: IKEv1 Main Mode Security Association negotiation

Figure 6 shows the Diffie-Hellman exchange phase. The Key Exchange payload carries the public Diffie-Hellman values required to generate the shared secret key between both peers. The Nonce payload adds randomness to the generated keys, increasing security.

The Key Exchange payload is therefore responsible for securely generating a common secret without transmitting the secret itself over the network.

47	17:24:45.833812	1.1.1.1	2.2.2.2	ISAKMP	390	Identity Protection (Main Mode)
48	17:24:46.001470	2.2.2.2	1.1.1.1	ISAKMP	410	Identity Protection (Main Mode)
49	17:24:46.265562	1.1.1.1	2.2.2.2	ISAKMP	118	Identity Protection (Main Mode)
50	17:24:46.446131	2.2.2.2	1.1.1.1	ISAKMP	118	Identity Protection (Main Mode)

```

> Frame 47: Packet, 390 bytes on wire (3120 bits), 390 bytes captured (3120 bits) on interface -, id 0
> Ethernet II, Src: 0c:d6:9b:f8:00:00 (0c:d6:9b:f8:00:00), Dst: ca:01:08:63:00:00 (ca:01:08:63:00:00)
> Internet Protocol Version 4, Src: 1.1.1.1, Dst: 2.2.2.2
> User Datagram Protocol, Src Port: 500, Dst Port: 500
> Internet Security Association and Key Management Protocol
  Initiator SPI: 77ece468cdcf914d
  Responder SPI: c2f86656590ef065
  Next payload: Key Exchange (4)
  > Version: 1.0
  Exchange type: Identity Protection (Main Mode) (2)
  > Flags: 0x00
  Message ID: 0x00000000
  Length: 348
  > Payload: Key Exchange (4)
  > Payload: Nonce (10)
  > Payload: Vendor ID (13) : RFC 3706 DPD (Dead Peer Detection)
  > Payload: Vendor ID (13) : Unknown Vendor ID
  > Payload: Vendor ID (13) : XAUTH
  > Payload: NAT-D (RFC 3947) (20)
  > Payload: NAT-D (RFC 3947) (20)

```

Figure 6: Diffie-Hellman Key Exchange during IKE Phase 1

Figure 7 shows the final Main Mode exchanges where identity information and authentication data are transmitted in encrypted form. At this stage, the ISAKMP SA has already been established.

49	17:24:46.265562	1.1.1.1	2.2.2.2	ISAKMP	118	Identity Protection (Main Mode)
50	17:24:46.446131	2.2.2.2	1.1.1.1	ISAKMP	118	Identity Protection (Main Mode)
51	17:24:46.597323	1.1.1.1	2.2.2.2	ISAKMP	214	Quick Mode
52	17:24:46.859881	2.2.2.2	1.1.1.1	ISAKMP	214	Quick Mode

```

> Frame 49: Packet, 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface -, id 0
> Ethernet II, Src: 0c:d6:9b:f8:00:00 (0c:d6:9b:f8:00:00), Dst: ca:01:08:63:00:00 (ca:01:08:63:00:00)
> Internet Protocol Version 4, Src: 1.1.1.1, Dst: 2.2.2.2
> User Datagram Protocol, Src Port: 500, Dst Port: 500
> Internet Security Association and Key Management Protocol
  Initiator SPI: 77ece468cdcf914d
  Responder SPI: c2f86656590ef065
  Next payload: Identification (5)
  > Version: 1.0
  Exchange type: Identity Protection (Main Mode) (2)
  > Flags: 0x01
  Message ID: 0x00000000
  Length: 76
  Encrypted Data (48 bytes)

```

Figure 7: Identity protection during IKE Phase 1

2.4.3 IKE Phase 2 – Quick Mode

IKE Phase 2 negotiates the IPSec Security Associations used to protect data traffic.

Figure 8 shows the Quick Mode exchange responsible for negotiating IPSec parameters such as AH/ESP protocols, encryption algorithms, integrity algorithms, and Security Associations.

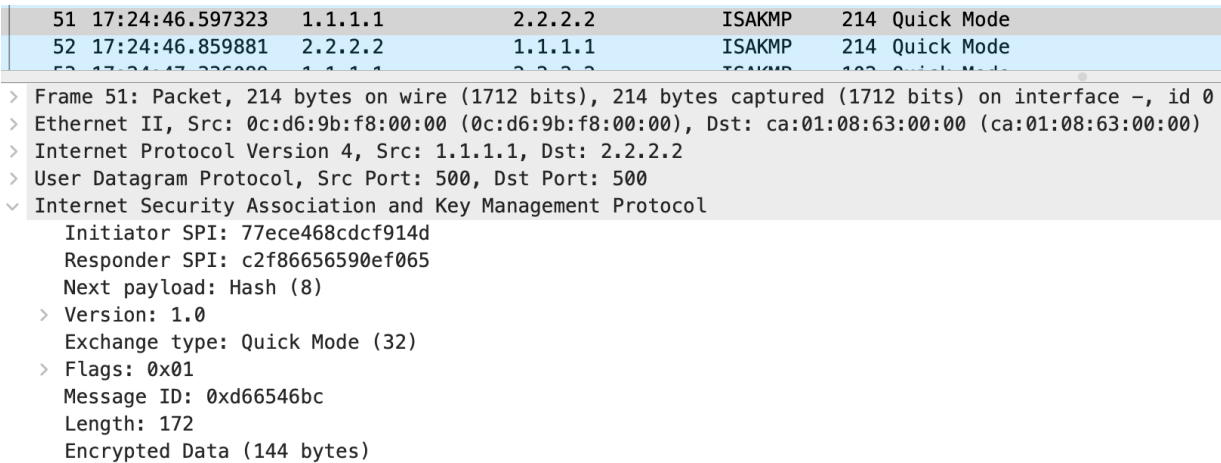


Figure 8: IKE Phase 2 Quick Mode negotiation

The router outputs confirmed the successful establishment of the tunnel. The ISAKMP state reached QM_IDLE, indicating that Phase 1 and Phase 2 negotiations completed successfully.

```
Router#show crypto isakmp sa
```

dst	src	state	conn-id	status
1.1.1.1	2.2.2.2	QM_IDLE	1002	ACTIVE
2.2.2.2	1.1.1.1	QM_IDLE	1003	ACTIVE

The IPSec Security Associations were also verified using:

```
show crypto ipsec sa
```

The output confirmed active inbound and outbound AH Security Associations.

2.5 Negotiation Failure Analysis

Different cryptographic parameters were intentionally configured between peers in order to analyze negotiation failures.

Figure 9 shows repeated Main Mode and Quick Mode negotiations. Since the peers could not agree on compatible IPSec parameters, the Security Associations were not successfully established.

As a result, ICMP communication failed and no encrypted traffic was exchanged through the tunnel.

151	12:09:26.242043	1.1.1.1	2.2.2.2	ISAKMP	118	Informational
152	12:09:26.288903	1.1.1.1	2.2.2.2	ISAKMP	118	Informational
153	12:09:26.493217	1.1.1.1	2.2.2.2	ISAKMP	134	Informational
154	12:09:27.316934	1.1.1.1	2.2.2.2	ISAKMP	210	Identity Protection (Main Mode)
155	12:09:27.443088	2.2.2.2	1.1.1.1	ISAKMP	150	Identity Protection (Main Mode)
156	12:09:27.652128	1.1.1.1	2.2.2.2	ISAKMP	390	Identity Protection (Main Mode)
157	12:09:27.886886	2.2.2.2	1.1.1.1	ISAKMP	410	Identity Protection (Main Mode)
158	12:09:28.100810	1.1.1.1	2.2.2.2	ISAKMP	150	Identity Protection (Main Mode)
160	12:09:28.255179	2.2.2.2	1.1.1.1	ISAKMP	118	Identity Protection (Main Mode)
162	12:09:28.672144	1.1.1.1	2.2.2.2	ISAKMP	214	Quick Mode
163	12:09:28.926799	2.2.2.2	1.1.1.1	ISAKMP	134	Informational
164	12:09:29.855434	2.2.2.2	1.1.1.1	ISAKMP	210	Identity Protection (Main Mode)
166	12:09:30.042589	1.1.1.1	2.2.2.2	ISAKMP	150	Identity Protection (Main Mode)
167	12:09:30.176062	2.2.2.2	1.1.1.1	ISAKMP	390	Identity Protection (Main Mode)
169	12:09:30.341166	1.1.1.1	2.2.2.2	ISAKMP	410	Identity Protection (Main Mode)
170	12:09:30.892237	2.2.2.2	1.1.1.1	ISAKMP	118	Identity Protection (Main Mode)
171	12:09:31.046912	1.1.1.1	2.2.2.2	ISAKMP	118	Identity Protection (Main Mode)
172	12:09:31.297021	2.2.2.2	1.1.1.1	ISAKMP	230	Quick Mode
173	12:09:31.401622	1.1.1.1	2.2.2.2	ISAKMP	134	Informational
185	12:10:00.474533	2.2.2.2	1.1.1.1	ISAKMP	230	Quick Mode
186	12:10:00.568552	1.1.1.1	2.2.2.2	ISAKMP	134	Informational
191	12:10:13.892447	1.1.1.1	2.2.2.2	ISAKMP	214	Quick Mode
192	12:10:13.959083	2.2.2.2	1.1.1.1	ISAKMP	134	Informational
211	12:10:58.901471	2.2.2.2	1.1.1.1	ISAKMP	230	Quick Mode
212	12:10:59.105788	1.1.1.1	2.2.2.2	ISAKMP	134	Informational
213	12:10:59.466884	1.1.1.1	2.2.2.2	ISAKMP	214	Quick Mode
214	12:10:59.621211	2.2.2.2	1.1.1.1	ISAKMP	134	Informational

Figure 9: Failed IPsec negotiation due to parameter mismatch

2.6 ESP Tunnel Operation

The transform set was later changed to ESP using the command:

```
crypto ipsec transform-set myTSet esp-aes esp-sha-hmac
```

ESP provides confidentiality in addition to authentication and integrity.

Figure 10 shows an ESP-protected packet. Unlike AH, only the ESP header remains visible, while the original payload is encrypted and cannot be inspected in Wireshark.

No.	Time	Source	Destination	Protocol	Leng	Info
2	12:30:08.264215	2.2.2.2	1.1.1.1	ESP	166	ESP (SPI=0x6e0b4bc2)
> Frame 2: Packet, 166 bytes on wire (1328 bits), 166 bytes captured (1328 bits) on interface -, id 0 > Ethernet II, Src: ca:01:08:63:00:00 (ca:01:08:63:00:00), Dst: 0c:d6:9b:f8:00:00 (0c:d6:9b:f8:00:00) > Internet Protocol Version 4, Src: 2.2.2.2, Dst: 1.1.1.1 > Encapsulating Security Payload ESP SPI: 0x6e0b4bc2 (1846234050) ESP Sequence: 21						

Figure 10: ESP protected packet

Figure 11 shows ESP packets exchanged together with OSPF Hello packets during tunnel operation.

```

→ 114 17:25:12.315783 192.168.1.100 192.168.4.100 ICMP 158 Echo (ping) request id=0x0002, seq=1/256, ttl=253 (reply in 115)
← 115 17:25:12.369215 192.168.4.100 192.168.1.100 ICMP 158 Echo (ping) reply id=0x0002, seq=1/256, ttl=253 (request in 114)
→ 116 17:25:12.414193 192.168.1.100 192.168.4.100 ICMP 158 Echo (ping) request id=0x0002, seq=2/512, ttl=253 (reply in 117)
> Frame 114: Packet, 158 bytes on wire (1264 bits), 158 bytes captured (1264 bits) on interface -, id 0
> Ethernet II, Src: 0c:d6:9b:f8:00:00 (0c:d6:9b:f8:00:00), Dst: ca:01:08:63:00:00 (ca:01:08:63:00:00)
> Internet Protocol Version 4, Src: 1.1.1.1, Dst: 2.2.2.2
  Authentication Header
    Next header: IPsec (4)
    Length: 4 (24 bytes)
    Reserved: 0000
    AH SPI: 0x2eb7d13b
    AH Sequence: 13
    AH ICV: 068fb9aecf6890bd68cfb25b
> Internet Protocol Version 4, Src: 192.168.1.100, Dst: 192.168.4.100
  Internet Control Message Protocol
    Type: Echo (ping) request (8)
    Code: 0
    Checksum: 0x8c21 [correct]
    [Checksum Status: Good]
    Identifier (BE): 2 (0x0002)
    Identifier (LE): 512 (0x0200)
    Sequence Number (BE): 1 (0x0001)
    Sequence Number (LE): 256 (0x0100)
    [Response frame: 115]
  Data (72 bytes)

```

Figure 11: ESP traffic and OSPF exchanges

Compared to AH, ESP provides confidentiality by encrypting the payload, while AH only guarantees integrity and authentication. In AH mode, the original payload remains visible in Wireshark, whereas in ESP mode only the ESP header can be inspected and the payload contents remain encrypted.

2.7 Firewall Requirements for IPsec

In order for an IPsec tunnel to be successfully established through a firewall, the firewall must allow the protocols and ports used by IKE and IPsec.

The following traffic must be permitted:

- UDP port 500 (ISAKMP/IKE)
- ESP protocol (IP protocol 50)
- AH protocol (IP protocol 51), when AH is used

If NAT traversal is used, UDP port 4500 must also be allowed.

Blocking any of these protocols prevents successful establishment of the IPsec tunnel.

3 IKEv1 with Digital Signatures

3.1 Introduction

This section describes the configuration and analysis of IPsec VPN using IKEv1 with digital certificates (PKI-based authentication using SCEP). The objective is to replace pre-shared keys with RSA signatures issued by a Certification Authority (CA).

3.2 Topology Overview

RA operates as the Certification Authority (CA) and NTP server. R1 and R2 establish an IPsec tunnel authenticated using RSA signatures and certificates obtained through SCEP enrollment.

3.3 PKI Installation Verification

The Certification Authority (CA) and router certificates were verified using IOS show commands.

3.3.1 CA Configuration and Certificate (RA)

The CA is configured on router RA and operates in automatic enrollment mode.

```
RA#show crypto pki server
Certificate Server myCA:
  Status: enabled
  State: enabled
  Server's configuration is locked (enter "shut" to unlock it)
  Issuer name: cn=saarCA
  CA cert fingerprint: 46CB9C10 CC370311 8CFCB545 E4DD2F58
  Granting mode is: auto
  Last certificate issued serial number (hex): 1
  CA certificate expiration timer: 16:17:23 UTC May 14 2029
  CRL NextUpdate timer: 22:17:23 UTC May 15 2026
  Current primary storage dir: nvram:
  Database Level: Minimum - no cert data written to storage
```

The CA certificate installed on RA confirms that RA is acting as a trusted root authority:

```
RA#show crypto ca certificate
CA Certificate
  Status: Available
  Certificate Serial Number (hex): 01
  Certificate Usage: Signature
  Issuer:
    cn=saarCA
  Subject:
```

```
cn=saarCA
Validity Date:
  start date: 16:17:23 UTC May 15 2026
  end   date: 16:17:23 UTC May 14 2029
Associated Trustpoints: myCA
Storage: nvram:saarCA#1CA.cer
```

The detailed certificate view confirms RSA key usage and X.509 extensions:

```
RA#show crypto ca certificate verbose
CA Certificate
Status: Available
Version: 3
Certificate Serial Number (hex): 01
Certificate Usage: Signature
Issuer:
  cn=saarCA
Subject:
  cn=saarCA
Validity Date:
  start date: 16:17:23 UTC May 15 2026
  end   date: 16:17:23 UTC May 14 2029
Subject Key Info:
  Public Key Algorithm: rsaEncryption
  RSA Public Key: (1024 bit)
Signature Algorithm: MD5 with RSA Encryption
Fingerprint MD5: 46CB9C10 CC370311 8CFCB545 E4DD2F58
Fingerprint SHA1: 0B6BF47E F0932FC1 54A33FA8 4C1F48D1 80083958
X509v3 extensions:
  X509v3 Key Usage: 86000000
    Digital Signature
    Key Cert Sign
    CRL Signature
  X509v3 Subject Key ID: 42BD5391 0D07D957 6CC26630 AABA09E4 A93162E2
  X509v3 Basic Constraints:
    CA: TRUE
  X509v3 Authority Key ID: 42BD5391 0D07D957 6CC26630 AABA09E4 A93162E2
  Authority Info Access:
Associated Trustpoints: myCA
Storage: nvram:saarCA#1CA.cer
```

3.3.2 Router Certificates (R1)

R1 successfully enrolled with the CA and received both the CA certificate and its own identity certificate:

```
R1#show crypto pki certificates
```

Certificate

Status: Available
Certificate Serial Number (hex): 03
Certificate Usage: General Purpose
Issuer:
 cn=saarCA
Subject:
 Name: R1
 Serial Number: 9S08DYF8WAKSOLHEG3ZVL
 hostname=R1+serialNumber=9S08DYF8WAKSOLHEG3ZVL
Validity Date:
 start date: 16:33:16 UTC May 15 2026
 end date: 16:33:16 UTC May 15 2027
Associated Trustpoints: myTrustpoint

The CA certificate is also stored locally to validate the trust chain:

CA Certificate

Status: Available
Certificate Serial Number (hex): 01
Certificate Usage: Signature
Issuer:
 cn=saarCA
Subject:
 cn=saarCA
Validity Date:
 start date: 16:17:23 UTC May 15 2026
 end date: 16:17:23 UTC May 14 2029
Associated Trustpoints: myTrustpoint

3.3.3 RSA Public Keys (R1 and R2)

The RSA public keys generated on R1 and R2 are extracted using the following IOS command:

```
show crypto key mypub rsa
```

R2 RSA Public Key:

```
R2#show crypto key mypub rsa
% Key pair was generated at: 16:32:05 UTC May 15 2026
Key name: R2
Key type: RSA KEYS
Storage Device: private-config
Usage: General Purpose Key
Key is not exportable.
Key Data:
```

```
30819F30 OD06092A 864886F7 OD010101 05000381 8D003081 89028181 00C337FE
B0DAEE4D 444B0AF7 A19A73B3 2D2FCD5E 27E09095 3017E393 2F27F7AA 581894E6
5E7F50B3 0454ACF2 FD1E8E24 F91125DD 6D611D93 611D70AD 36C59F6F EAEE003C
7704B6DB 41D9A165 27252864 17170DA2 F38A68D4 46C91133 36DBCBC5 B09AD90E
0F05531B F7ABB4A5 0B954C0F 389C2195 3DE377FF 7FE63FB3 B02DF189 35020301
0001
```

% Key pair was generated at: 16:57:46 UTC May 16 2026

Key name: R2.server

Key type: RSA KEYS

Temporary key

Usage: Encryption Key

Key is not exportable.

Key Data:

```
307C300D 06092A86 4886F70D 01010105 00036B00 30680261 00A38B5F B33F29E4
E5E30AFB 512084C9 0A89E015 BCA1569A C10C686B B79BD8EC C3C25CAD 6EE74C8F
C891E5FF 1B042503 710EC940 65712523 377F2776 77CDC4C0 D7D61754 57C88063
19BF191B F48130CC 327EF733 1F877D95 5CB74450 AEEF7E99 45020301 0001
```

R1 RSA Public Key:

R1#show crypto key mypub rsa

% Key pair was generated at: 16:31:43 UTC May 15 2026

Key name: R1

Key type: RSA KEYS

Storage Device: private-config

Usage: General Purpose Key

Key is not exportable.

Key Data:

```
30819F30 OD06092A 864886F7 OD010101 05000381 8D003081 89028181 00DF87BB
11BD33EE C4E9DF4A C2FF2CCB 2ECCEFF9 713AAA45 633ADE4A C188109E 88680D67
FBOA2F20 67257007 79DB3616 75D5459A 4898DE69 E74E80BA 77C993E8 A4B70D41
00D1D2FB E37BF0E3 89C8BEEE 9830712E 8A4C13DC E30C5C72 D72FEDF2 A221F6DB
9A1A1B8F 9A69BC61 3700CA1B B66DB044 72F24609 E0C1FE8D 24D1B3DA DD020301
0001
```

% Key pair was generated at: 16:57:42 UTC May 16 2026

Key name: R1.server

Key type: RSA KEYS

Temporary key

Usage: Encryption Key

Key is not exportable.

Key Data:

```
307C300D 06092A86 4886F70D 01010105 00036B00 30680261 0095CC14 73EE9BA8
1B0A40A6 E3EE90C1 BC2FAADF 2D532270 160D9039 C19C40CA 6A4E54B5 BAB8892C
D266AE9A 53210F58 876755F6 381C44B6 5D71F57F 35705BC5 3E8D9746 61833A6A
DB5E740B 8A22A486 83EE7AFA ACAB3798 27D03FC2 7C6EA875 FD020301 0001
```


Both routers generate 1024-bit RSA key pairs used for digital signature authentication during IKEv1 Phase 1.

3.4 SCEP Enrollment Analysis

SCEP (Simple Certificate Enrollment Protocol) is used to automate certificate distribution between routers and the CA using HTTP.

The enrollment process consists of:

- CA certificate request
- CA certificate response
- Router certificate request
- Router certificate issuance

3.4.1 CA Certificate Retrieval

Initially, R1 requests the CA certificate using the SCEP GetCACert operation over HTTP.

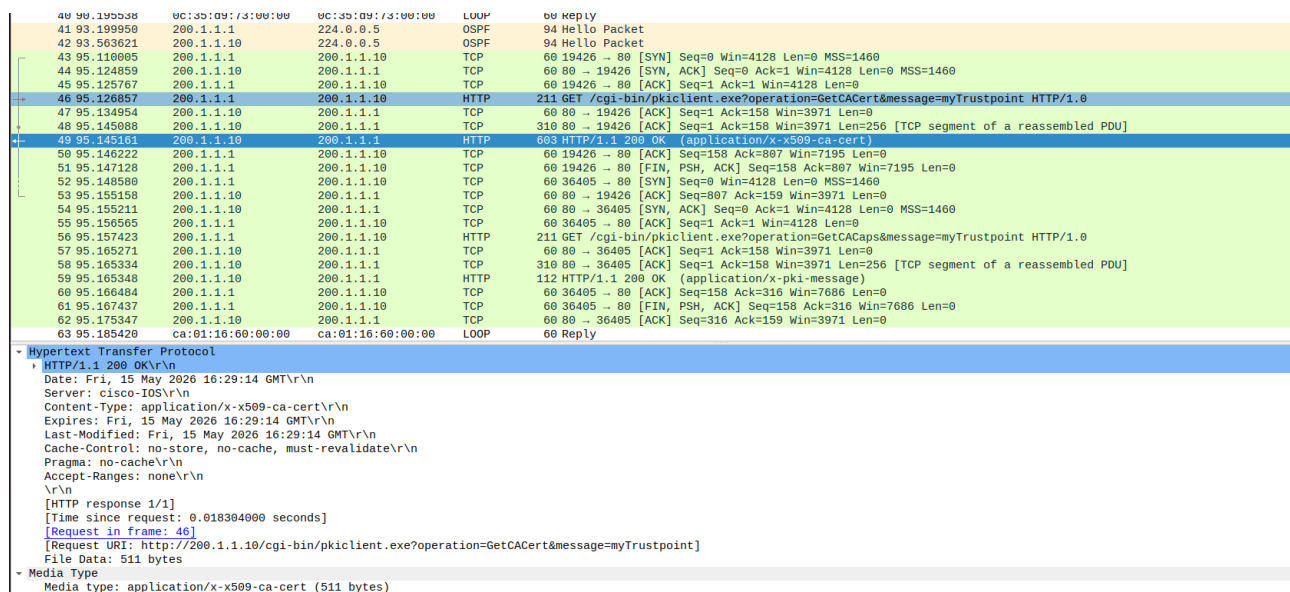


Figure 12: SCEP GetCACert request and CA certificate response

The CA replies with an HTTP 200 OK message containing an X.509 CA certificate with media type `application/x-x509-ca-cert`. This certificate allows R1 to establish trust with the Certification Authority before certificate enrollment.

3.4.2 Certificate Enrollment Request

After establishing trust with the CA, R1 performs certificate enrollment using the SCEP `PKIOperation` request.

169	330.534670	200.1.1.10	224.0.0.5	OSPF	94 Hello Packet
170	333.552158	ca:01:16:00:00:00	CDP/VTP/DTP/PagP/UD...	CDP	349 Device ID: RA Port ID: FastEthernet0/0
171	335.184788	ca:01:16:00:00:00	ca:01:16:00:00:00	LOOP	60 Reply
172	335.756961	200.1.1.1	224.0.0.5	OSPF	94 Hello Packet
173	336.711505	200.1.1.1	200.1.1.10	TCP	60 55472 → 80 [SYN] Seq=0 Win=4128 Len=0 MSS=1460
174	336.730711	200.1.1.10	200.1.1.1	TCP	60 80 → 55472 [SYN, ACK] Seq=0 Ack=1 Win=4128 Len=0 MSS=1460
175	336.731861	200.1.1.1	200.1.1.10	TCP	60 55472 → 80 [ACK] Seq=1 Ack=1 Win=4128 Len=0
176	336.732806	200.1.1.1	200.1.1.10	TCP	1514 55472 → 80 [ACK] Seq=1 Ack=1 Win=4128 Len=1460 [TCP segment of a reassembled PDU]
177	336.740658	200.1.1.10	200.1.1.1	TCP	60 80 → 55472 [ACK] Seq=1 Ack=1461 Win=2668 Len=0
178	336.740715	200.1.1.10	200.1.1.1	TCP	60 [TCP Window Update] 80 → 55472 [ACK] Seq=1 Ack=1461 Win=4128 Len=0
179	336.743055	200.1.1.1	200.1.1.10	HTTP	907 GET /cgi-bin/pkiclient.exe?operation=PKIOperation&message=MIIF7QYJKoZIhvcNAQcCoIIF3jCCBdoCAQExCzAJBgUrDgMCGGUAMIICvgYJKoZI%0Ahvvc
180	336.750752	200.1.1.10	200.1.1.1	TCP	310 80 → 55472 [ACK] Seq=1 Ack=2314 Win=3275 Len=256 [TCP segment of a reassembled PDU]
181	336.901694	200.1.1.10	200.1.1.1	TCP	1259 80 → 55472 [ACK] Seq=257 Ack=2314 Win=3275 Len=1205 [TCP segment of a reassembled PDU]
182	336.904198	200.1.1.1	200.1.1.10	TCP	60 55472 → 80 [ACK] Seq=2314 Ack=1462 Win=6539 Len=0
183	336.904245	200.1.1.1	200.1.1.10	TCP	60 [TCP Window Update] 55472 → 80 [ACK] Seq=2314 Ack=1462 Win=8000 Len=0
184	336.911797	200.1.1.10	200.1.1.1	TCP	166 80 → 55472 [ACK] Seq=1462 Ack=2314 Win=3275 Len=112 [TCP segment of a reassembled PDU]
185	336.911885	200.1.1.10	200.1.1.1	HTTP	71 HTTP/1.1 200 OK (application/x-pki-message)
186	336.916086	200.1.1.1	200.1.1.10	TCP	60 55472 → 80 [ACK] Seq=2314 Ack=1592 Win=7871 Len=0
187	336.916127	200.1.1.1	200.1.1.10	TCP	60 55472 → 80 [FIN, PSH, ACK] Seq=2314 Ack=1592 Win=7871 Len=0
188	336.921879	200.1.1.10	200.1.1.1	TCP	60 80 → 55472 [ACK] Seq=1592 Ack=2315 Win=3275 Len=0
189	339.758848	200.1.1.10	224.0.0.5	OSPF	94 Hello Packet
190	340.502206	0c:35:d9:73:00:00	0c:35:d9:73:00:00	LOOP	60 Reply
191	344.963580	200.1.1.1	224.0.0.5	OSPF	94 Hello Packet
192	345.184696	ca:01:16:00:00:00	ca:01:16:00:00:00	LOOP	60 Reply

Response Phrase: OK	
Date: Fri, 15 May 2026 16:33:16 GMT\r\n	
Server: cisco-IOS\r\n	
Content-Type: application/x-pki-message\r\n	
Expires: Fri, 15 May 2026 16:33:16 GMT\r\n	
Last-Modified: Fri, 15 May 2026 16:33:16 GMT\r\n	
Cache-Control: no-store, no-cache, must-revalidate\r\n	
Pragma: no-cache\r\n	
Accept-Ranges: none\r\n	
\r\n	
[HTTP response 1/1]	
[Time since request: 0.168830000 seconds]	
[Request in frame: 179]	
[Request URI [truncated]: http://200.1.1.10/cgi-bin/pkiclient.exe?operation=PKIOperation&message=MIIF7QYJKoZIhvcNAQcCoIIF3jCCBdoCAQExCzAJBgUrDgMCGGUAMIICvgYJKoZI%0Ahvvc]	
File Data: 1297 bytes	

Media Type	
Media type: application/x-pki-message (1297 bytes)	

Figure 13: SCEP PKIOperation certificate enrollment request

The HTTP request contains a PKCS#10 certificate signing request (CSR) encoded in the message field. The CSR includes the router identity and its RSA public key, allowing the CA to generate a signed X.509 certificate for R1.

The captured HTTP messages show certificate exchange between R1 and the CA. The content-type field indicates certificate payload transfer (.crt).

Using Wireshark, the packet containing the issued certificate was identified by filtering HTTP traffic and locating the response message with a **Media Type** indicating a certificate object. The packet was then selected, and the certificate data was extracted by right-clicking on the relevant field and choosing *Export Packet Bytes*. The file was saved with a .crt extension, which allows it to be decoded as an X.509 certificate.

packet.crt	
saarCA	
Identity: saarCA	
Verified by: saarCA	
Expires: 14/05/2029	
Details	
Subject Name	
CN (Common Name): saarCA	
Issuer Name	
CN (Common Name): saarCA	
Issued Certificate	
Version: 3	
Serial Number: 01	
Not Valid Before: 2026-05-15	
Not Valid After: 2029-05-14	
Certificate Fingerprints	
SHA1: 0B 6B F4 7E F0 93 2F C1 54 A3 3F A8 4C 1F 48 D1 80 08 39 58	
MD5: 46 CB 9C 10 CC 37 03 11 8C FC B5 45 E4 DD 2F 58	
Public Key Info	
Key Algorithm: RSA	
Key Parameters: 05 00	
Key Size: 1024	
Key SHA1 Fingerprint: BD 39 CD 1B 6D A4 0C CF 6A 75 25 4D A0 C5 53 AE 85 8D 41 C9	
Public Key: 30 81 89 02 81 81 00 A3 6A 1F EA 2A 1C B5 81 90 E4 BF CA 59 FD D2 76 17 CF 05 D2 AD 94 AD 74 EA BC 27 C2 45 7C 45 6D 2F C0 5E D9 DC B2 30 B8 51 67 9A 20 30 40 55 A2 9C D0 F8 9C 75 46 D2 0A 07 70 82 66 58 58 3D 5E 2E 18 FE FB 94 81 ED 2F A0 95 67 EB A4 97 AA 82 BC 13 CD C3 58 9E 22 F9 DC ED 01 75 92 5A EB 61 25 E1 2A 38 33 83 A3 47 76 59 8F 0F 02 03 01 00 01	
Basic Constraints	
Certificate Authority: Yes	
Max Path Length: Unlimited	
Critical: Yes	
Key Usage	
Usages: Digital signature ↵	
Certificate signature ↵	
Revocation list signature	
Critical: Yes	
Extension	
Identifier: 2.5.29.35	
Value: 30 16 80 14 42 BD 53 91 0D 07 D9 57 6C C2 66 30 AA BA 09 E4 A9 31 62 E2	
Critical: No	
Subject Key Identifier	
Key Identifier: 42 BD 53 91 0D 07 D9 57 6C C2 66 30 AA BA 09 E4 A9 31 62 E2	
Critical: No	
Signature	

Figure 14: HTTP packet carrying the issued X.509 certificate (.crt file export in Wireshark)

The exported certificate represents the CA-signed identity certificate for R1. It contains the subject information, the public key corresponding to R1, and the digital signature generated by the CA, which validates its authenticity within the PKI infrastructure.

3.5 Encrypted ICMP Traffic Analysis

After configuring IPsec with RSA-signature authentication, connectivity between PC1 and PC2 was successfully verified using ICMP echo requests (ping).

Traffic was captured on the public network between R1 and R2 while the ping operation was performed.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:35:d9:73:00:00	0c:35:d9:73:00:00	LOOP	60	Reply
2	3.889135	200.1.1.10	224.0.0.5	OSPF	94	Hello Packet
3	5.482695	ca:01:16:60:00:00	ca:01:16:60:00:00	LOOP	60	Reply
4	6.320045	2.2.2.2	1.1.1.1	ESP	166	ESP (SPI=0xb87d2aab)
5	7.009428	1.1.1.1	2.2.2.2	ESP	166	ESP (SPI=0x147d646a)
6	7.269011	200.1.1.1	224.0.0.5	OSPF	94	Hello Packet
7	8.171183	2.2.2.2	1.1.1.1	ESP	182	ESP (SPI=0xb87d2aab)
8	8.188890	1.1.1.1	2.2.2.2	ESP	182	ESP (SPI=0x147d646a)
9	8.221515	2.2.2.2	1.1.1.1	ESP	182	ESP (SPI=0xb87d2aab)
10	8.239182	1.1.1.1	2.2.2.2	ESP	182	ESP (SPI=0x147d646a)
11	8.282182	2.2.2.2	1.1.1.1	ESP	182	ESP (SPI=0xb87d2aab)
12	8.299668	1.1.1.1	2.2.2.2	ESP	182	ESP (SPI=0x147d646a)
13	8.322552	2.2.2.2	1.1.1.1	ESP	182	ESP (SPI=0xb87d2aab)
14	8.339979	1.1.1.1	2.2.2.2	ESP	182	ESP (SPI=0x147d646a)
15	8.362850	2.2.2.2	1.1.1.1	ESP	182	ESP (SPI=0xb87d2aab)
16	8.380385	1.1.1.1	2.2.2.2	ESP	182	ESP (SPI=0x147d646a)
17	10.028466	0c:35:d9:73:00:00	0c:35:d9:73:00:00	LOOP	60	Reply
18	13.658516	200.1.1.10	224.0.0.5	OSPF	94	Hello Packet
19	15.485532	ca:01:16:60:00:00	ca:01:16:60:00:00	LOOP	60	Reply
20	15.847636	2.2.2.2	1.1.1.1	ESP	166	ESP (SPI=0xb87d2aab)
21	16.864505	200.1.1.1	224.0.0.5	OSPF	94	Hello Packet
22	17.007888	1.1.1.1	2.2.2.2	ESP	166	ESP (SPI=0x147d646a)
23	20.042916	0c:35:d9:73:00:00	0c:35:d9:73:00:00	LOOP	60	Reply


```

▶ Frame 7: 182 bytes on wire (1456 bits), 182 bytes captured (1456 bits) on interface -, id 0
▶ Ethernet II, Src: ca:01:16:60:00:00 (ca:01:16:60:00:00), Dst: 0c:35:d9:73:00:00 (0c:35:d9:73:00:00)
▼ Internet Protocol Version 4, Src: 2.2.2.2, Dst: 1.1.1.1
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 168
    Identification: 0x011d (285)
  ▶ Flags: 0x00
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 254
    Protocol: Encap Security Payload (50)
    Header Checksum: 0xb501 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 2.2.2.2
    Destination Address: 1.1.1.1
  ▼ Encapsulating Security Payload
    ESP SPI: 0xb87d2aab (3095210667)
    ESP Sequence: 61

```

Figure 15: Encrypted IPsec traffic observed between R1 and R2

The Wireshark capture shows that the ICMP packets exchanged between PC1 and PC2 are not visible in plaintext on the public network. Instead, the packets appear encapsulated using ESP (Encapsulating Security Payload).

The ESP packets contain encrypted payload data, preventing direct inspection of the original ICMP echo request and echo reply contents. Only the external IP header remains visible, showing communication between the tunnel endpoints (R1 and R2).

This confirms that IPsec is correctly providing:

- **Confidentiality** through payload encryption
- **Integrity protection** through ESP authentication
- **Secure tunnel encapsulation** between the two sites

The observed ESP traffic corresponds to the IPsec transform-set configuration:

```
crypto ipsec transform-set myTSet esp-aes esp-sha-hmac
```

AES is used for encryption while SHA-HMAC is used for integrity verification.

3.6 ISAKMP and SA Re-establishment Analysis

To analyze tunnel re-establishment, the IPSec Security Associations (SAs) were manually cleared using:

```
clear crypto session
```

While capturing traffic on the public network, new ISAKMP exchanges were observed as the routers negotiated fresh IPSec Security Associations.

1 0.000000	1.1.1.1	2.2.2.2	ESP	166 ESP (SPI=0x37bfac13)
2 1.087267	2.2.2.2	1.1.1.1	ESP	166 ESP (SPI=0x3aa0a0ef)
3 1.968939	200.1.1.10	224.0.0.5	OSPF	94 Hello Packet
4 4.571026	200.1.1.1	224.0.0.5	OSPF	94 Hello Packet
5 5.111299	0c:35:d9:73:00:00	0c:35:d9:73:00:00	LOOP	60 Reply
6 5.470172	ca:01:16:60:00:00	ca:01:16:60:00:00	LOOP	60 Reply
7 7.109808	ca:01:16:60:00:00	CDP/VTP/DTP/PagP/UD...	CDP	349 Device ID: RA Port ID: FastEthernet0/0
8 8.201714	1.1.1.1	2.2.2.2	ISAKMP	118 Informational
9 8.203333	1.1.1.1	2.2.2.2	ISAKMP	134 Informational
10 8.217971	2.2.2.2	1.1.1.1	ISAKMP	134 Informational
11 8.888812	1.1.1.1	2.2.2.2	ISAKMP	210 Identity Protection (Main Mode)
12 8.901438	2.2.2.2	1.1.1.1	ISAKMP	150 Identity Protection (Main Mode)
13 8.907525	1.1.1.1	2.2.2.2	ISAKMP	414 Identity Protection (Main Mode)
14 8.931954	2.2.2.2	1.1.1.1	ISAKMP	434 Identity Protection (Main Mode)
15 8.968150	1.1.1.1	2.2.2.2	ISAKMP	742 Identity Protection (Main Mode)
16 9.012585	2.2.2.2	1.1.1.1	ISAKMP	742 Identity Protection (Main Mode)
17 9.031615	1.1.1.1	2.2.2.2	ISAKMP	230 Quick Mode
18 9.054747	2.2.2.2	1.1.1.1	ISAKMP	230 Quick Mode
19 9.072084	1.1.1.1	2.2.2.2	ISAKMP	102 Quick Mode
20 9.073561	1.1.1.1	2.2.2.2	ESP	150 ESP (SPI=0x59df71ad)
21 9.084887	2.2.2.2	1.1.1.1	ESP	150 ESP (SPI=0x08155835)
22 9.087434	1.1.1.1	2.2.2.2	ESP	166 ESP (SPI=0x59df71ad)
23 9.105426	2.2.2.2	1.1.1.1	ESP	150 ESP (SPI=0x08155835)
24 9.105619	2.2.2.2	1.1.1.1	ESP	166 ESP (SPI=0x08155835)
25 9.109976	1.1.1.1	2.2.2.2	ESP	150 ESP (SPI=0x59df71ad)
26 9.110573	1.1.1.1	2.2.2.2	ESP	262 ESP (SPI=0x59df71ad)
27 9.125696	2.2.2.2	1.1.1.1	ESP	182 ESP (SPI=0x08155835)
28 9.127906	1.1.1.1	2.2.2.2	ESP	134 ESP (SPI=0x59df71ad)
29 9.128484	1.1.1.1	2.2.2.2	ESP	150 ESP (SPI=0x59df71ad)
30 9.145928	2.2.2.2	1.1.1.1	ESP	166 ESP (SPI=0x08155835)
31 9.145991	2.2.2.2	1.1.1.1	ESP	134 ESP (SPI=0x08155835)
32 9.148997	1.1.1.1	2.2.2.2	ESP	166 ESP (SPI=0x59df71ad)
33 11.649739	200.1.1.10	224.0.0.5	OSPF	94 Hello Packet
34 11.661379	1.1.1.1	2.2.2.2	ESP	150 ESP (SPI=0x59df71ad)
35 11.680534	2.2.2.2	1.1.1.1	ESP	150 ESP (SPI=0x08155835)
36 13.733941	1.1.1.1	2.2.2.2	ESP	182 ESP (SPI=0x59df71ad)
37 13.754607	2.2.2.2	1.1.1.1	ESP	182 ESP (SPI=0x08155835)
38 14.254637	200.1.1.1	224.0.0.5	OSPF	94 Hello Packet
39 15.160283	0c:35:d9:73:00:00	0c:35:d9:73:00:00	LOOP	60 Reply
40 15.468087	ca:01:16:60:00:00	ca:01:16:60:00:00	LOOP	60 Reply
41 16.248445	2.2.2.2	1.1.1.1	ESP	150 ESP (SPI=0x08155835)
42 16.264702	1.1.1.1	2.2.2.2	ESP	150 ESP (SPI=0x59df71ad)
43 18.238626	1.1.1.1	2.2.2.2	ESP	166 ESP (SPI=0x59df71ad)
44 18.409561	2.2.2.2	1.1.1.1	ESP	166 ESP (SPI=0x08155835)
45 20.827957	200.1.1.10	224.0.0.5	OSPF	94 Hello Packet

Figure 16: ISAKMP exchanges after clearing crypto sessions

After the existing SAs were deleted, R1 and R2 initiated a new IKEv1 Phase 1 negotiation using ISAKMP messages over UDP port 500.

The captured exchange shows the following operations:

- ISAKMP SA proposal negotiation
- Exchange of nonces and Diffie-Hellman values
- Certificate exchange between peers

- RSA digital signature authentication
- Establishment of a new ISAKMP Security Association
- Negotiation of IPsec Phase 2 parameters
- Creation of new IPsec SAs for ESP traffic

The certificate-based authentication process relies on the X.509 certificates previously obtained from the CA through SCEP enrollment.

During authentication, each router validates:

- The peer certificate authenticity
- The CA signature
- The RSA signature generated during IKE authentication

Successful validation allows the tunnel to be securely re-established without the use of pre-shared keys.

The re-established IPsec tunnel was confirmed by successful encrypted ICMP communication between PC1 and PC2 after the negotiation process completed.

3.7 Peer Public Key Storage Verification

After successful IKEv1 authentication and IPsec tunnel establishment, each router stores the public key information of the remote peer.

At R1, the public key of R2 can be viewed using:

```
show crypto key pubkey-chain rsa name R2
```

The output confirms that R1 has learned and stored the RSA public key associated with R2's certificate.

```
R1#sh crypto key pubkey-chain rsa name R2
```

```
Key name: R2
```

```
Subject name(X.500 DN name): hostname=R2+serialNumber=9C92ZG7WMXGLWHR7XWAK8
```

```
Key id: 11
```

```
Serial number: 02
```

```
Usage: Signature Key
```

```
Source: Certificate
```

```
Data:
```

```
30819F30 0D06092A 864886F7 0D010101 05000381 8D003081 89028181 00C337FE
B0DAEE4D 444B0AF7 A19A73B3 2D2FCD5E 27E09095 3017E393 2F27F7AA 581894E6
5E7F50B3 0454ACF2 FD1E8E24 F91125DD 6D611D93 611D70AD 36C59F6F EAEE003C
7704B6DB 41D9A165 27252864 17170DA2 F38A68D4 46C91133 36DBCBC5 B09AD90E
0F05531B F7ABB4A5 0B954C0F 389C2195 3DE377FF 7FE63FB3 B02DF189 35020301
0001
```

This public key is extracted from the X.509 certificate presented by R2 during the IKEv1 authentication process.

The stored public key allows R1 to:

- Verify digital signatures generated by R2
- Validate the identity of the remote peer
- Establish trusted IPsec communications

Similarly, R2 stores the public key of R1 after successful authentication.

The existence of the peer public key chain confirms that certificate-based authentication using RSA signatures was successfully completed and that both routers trust certificates issued by the Certification Authority.

3.8 SCEP Messages Observed After Router Restart

After stopping and restarting all routers, Wireshark captures were collected on the public network shortly after boot.

During startup, HTTP messages related to SCEP were again exchanged between the routers and the CA.

No.	Time	Source	Destination	Protocol	Length	Info
87	91.311649	2.2.2.2	1.1.1.1	ISAKMP	210	Identity Protection (Main Mode)
88	91.315619	1.1.1.1	2.2.2.2	ISAKMP	150	Identity Protection (Main Mode)
89	91.332219	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
90	91.350331	1.1.1.1	2.2.2.2	ISAKMP	434	Identity Protection (Main Mode)
91	91.382800	2.2.2.2	1.1.1.1	ISAKMP	774	Identity Protection (Main Mode)
92	91.406319	200.1.1.1	200.1.1.10	TCP	60	16204 → 80 [SYN] Seq=0 Win=4128 Len=0 MSS=1460
93	91.444908	200.1.1.10	200.1.1.1	TCP	60	80 → 16204 [SYN, ACK] Seq=0 Ack=1 Win=4128 Len=0 MSS=1460
94	91.445726	200.1.1.1	200.1.1.10	TCP	60	16204 → 80 [ACK] Seq=1 Ack=1 Win=4128 Len=0
95	91.446541	200.1.1.1	200.1.1.10	HTTP	211	GET /cgi-bin/pkiclient.exe?operation=GetCACaps&message=myTrustpoint HTTP/1.0
96	91.454995	200.1.1.10	200.1.1.1	TCP	60	80 → 16204 [ACK] Seq=1 Ack=158 Win=3971 Len=0
97	91.475169	200.1.1.10	200.1.1.1	TCP	310	80 → 16204 [ACK] Seq=1 Ack=158 Win=3971 Len=256 [TCP segment of a reassembled PDU]
98	91.475234	200.1.1.10	200.1.1.1	HTTP	112	HTTP/1.1 200 OK (application/x-pki-message)
99	91.476088	200.1.1.1	200.1.1.10	TCP	60	16204 → 80 [ACK] Seq=158 Ack=316 Win=7686 Len=0
100	91.476837	200.1.1.1	200.1.1.10	TCP	60	16204 → 80 [FIN, PSH, ACK] Seq=158 Ack=316 Win=7686 Len=0
101	91.485277	200.1.1.10	200.1.1.1	TCP	60	80 → 16204 [ACK] Seq=316 Ack=159 Win=3971 Len=0
102	91.587719	200.1.1.1	200.1.1.10	TCP	60	22215 → 80 [SYN] Seq=0 Win=4128 Len=0 MSS=1460
103	91.590602	200.1.1.10	200.1.1.1	TCP	60	80 → 22215 [SYN, ACK] Seq=0 Ack=1 Win=4128 Len=0 MSS=1460
104	91.596765	200.1.1.1	200.1.1.10	TCP	60	22215 → 80 [ACK] Seq=1 Ack=1 Win=4128 Len=0
105	91.597308	200.1.1.1	200.1.1.10	TCP	1514	22215 → 80 [ACK] Seq=1 Ack=1 Win=4128 Len=1460 [TCP segment of a reassembled PDU]
106	91.606136	200.1.1.10	200.1.1.1	TCP	60	80 → 22215 [ACK] Seq=1 Ack=1461 Win=2668 Len=0
107	91.606197	200.1.1.10	200.1.1.1	TCP	60	[TCP Window Update] 80 → 22215 [ACK] Seq=1 Ack=1461 Win=4128 Len=0
108	91.607833	200.1.1.1	200.1.1.10	HTTP	418	GET /cgi-bin/pkiclient.exe?operation=PKIOperation&message=MIIEoQYJKoZIhvcNAQCoIIEkjcCB14CAQExCzAJBgUrDgMCGgUAMIIIBIAYJ...
109	91.616244	200.1.1.10	200.1.1.1	TCP	310	80 → 22215 [ACK] Seq=1 Ack=1825 Win=3764 Len=256 [TCP segment of a reassembled PDU]
110	91.652295	1.1.1.1	2.2.2.2	ISAKMP	210	Identity Protection (Main Mode)
111	91.666685	2.2.2.2	1.1.1.1	ISAKMP	150	Identity Protection (Main Mode)
112	91.670489	1.1.1.1	2.2.2.2	ISAKMP	414	Identity Protection (Main Mode)
113	91.697817	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
114	91.729434	200.1.1.10	200.1.1.1	HTTP	3943	HTTP/1.1 200 OK (application/x-pki-message)
115	91.734258	1.1.1.1	2.2.2.2	ISAKMP	742	Identity Protection (Main Mode)
116	91.735353	200.1.1.1	200.1.1.10	TCP	60	22215 → 80 [ACK] Seq=1825 Ack=1247 Win=6755 Len=0
117	91.736353	200.1.1.1	200.1.1.10	TCP	60	22215 → 80 [FIN, PSH, ACK] Seq=1825 Ack=1247 Win=6755 Len=0
118	91.737934	200.1.1.10	200.1.1.1	TCP	60	80 → 22215 [ACK] Seq=1247 Ack=1826 Win=3764 Len=0
119	91.859141	1.1.1.1	2.2.2.2	ISAKMP	742	Identity Protection (Main Mode)
120	92.115710	2.2.2.2	1.1.1.1	ISAKMP	742	Identity Protection (Main Mode)
121	92.131452	1.1.1.1	2.2.2.2	ISAKMP	230	Quick Mode
122	92.146105	2.2.2.2	1.1.1.1	ISAKMP	230	Quick Mode
123	92.158653	1.1.1.1	2.2.2.2	ISAKMP	182	Quick Mode
124	92.159899	1.1.1.1	2.2.2.2	ESP	150	ESP (SPI=0xaa824a3a)
125	92.177822	2.2.2.2	1.1.1.1	ESP	150	ESP (SPI=0xaa824a3a)
126	92.178695	1.1.1.1	2.2.2.2	ESP	166	ESP (SPI=0xaa824a3a)
127	92.198041	2.2.2.2	1.1.1.1	ESP	150	ESP (SPI=0xaa824a3a)
128	92.198103	2.2.2.2	1.1.1.1	ESP	166	ESP (SPI=0xaa824a3a)
129	92.201255	1.1.1.1	2.2.2.2	ESP	150	ESP (SPI=0xaa824a3a)
130	92.201780	1.1.1.1	2.2.2.2	ESP	198	ESP (SPI=0xaa824a3a)
131	92.218377	2.2.2.2	1.1.1.1	ESP	198	ESP (SPI=0xaa824a3a)

Figure 17: SCEP-related HTTP traffic observed after router restart

The captured traffic includes HTTP requests and responses associated with certificate validation and certificate status retrieval operations.

These SCEP exchanges occur because the routers must re-establish trust information after reboot and verify the validity of the installed certificates before using them for IPsec authentication.

The observed messages serve several purposes:

- Verification that the CA is reachable
- Retrieval or validation of CA information
- Validation of certificate status
- Synchronization of trust-related information

This mechanism ensures that routers do not use invalid, expired, or revoked certificates during IKEv1 authentication.

The SCEP communication therefore plays an essential role in maintaining the integrity and trustworthiness of the PKI infrastructure used for IPsec VPN authentication.

3.9 Certificate Revocation and Authentication Failure

The certificates issued to R1 and R2 were revoked at the Certification Authority using:

```
RA#crypto pki server myCA revoke 02
RA#crypto pki server myCA revoke 03
```

After revocation, all routers were restarted and Wireshark captures were collected on the links between the routers and the CA.

When attempting communication between PC1 and PC2, the ping operation failed and the IPsec tunnel could not be established.

3.9.1 SCEP and Certificate Validation Traffic

After reboot, HTTP/SCEP-related messages were observed between the routers and the CA.

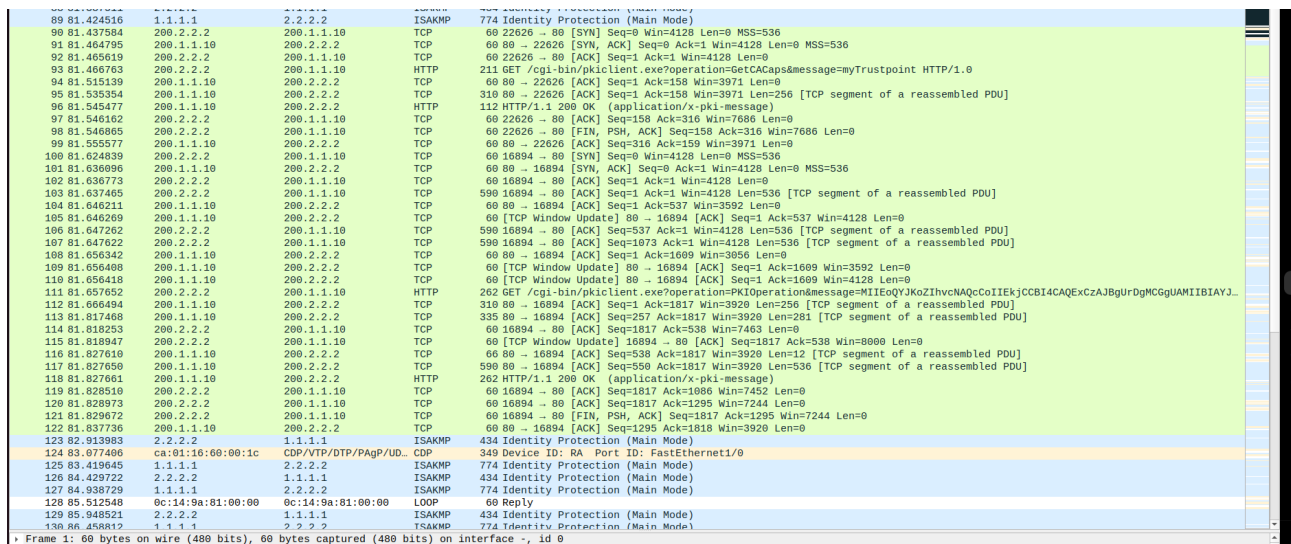


Figure 18: SCEP-related traffic observed after certificate revocation

These messages are associated with certificate validation and trust verification operations performed before IPsec authentication.

The routers contact the CA to verify the validity of certificates and retrieve updated revocation information.

3.9.2 Failed ISAKMP Negotiation

While the routers attempted to establish the IPsec tunnel, repeated ISAKMP exchanges were observed.

No.	Time	Source	Destination	Protocol	Length	Info
143	121.379416	1.1.1.1	2.2.2.2	ISAKMP	434	Identity Protection (Main Mode)
144	121.426688	2.2.2.2	1.1.1.1	ISAKMP	742	Identity Protection (Main Mode)
148	121.448431	200.1.1.1	200.1.1.10	HTTP	211	GET /cgi-bin/pkiclient.exe?operation=GetCACaps&message=myTrustpoint HTTP/1.0
151	121.457924	200.1.1.10	200.1.1.1	HTTP	112	HTTP/1.1 200 OK (application/x-pki-message)
161	121.579467	200.1.1.1	200.1.1.10	HTTP	410	GET /cgi-bin/pkiclient.exe?operation=PKIOperation&message=MIIEoQYJKoZIhvcNAQCoIIEkjCCB...
163	121.663616	1.1.1.1	2.2.2.2	ISAKMP	210	Identity Protection (Main Mode)
164	121.680646	2.2.2.2	1.1.1.1	ISAKMP	150	Identity Protection (Main Mode)
165	121.684412	1.1.1.1	2.2.2.2	ISAKMP	414	Identity Protection (Main Mode)
166	121.700732	200.1.1.10	200.1.1.1	HTTP	1091	HTTP/1.1 200 OK (application/x-pki-message)
170	121.710938	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
171	121.808940	1.1.1.1	2.2.2.2	ISAKMP	742	Identity Protection (Main Mode)
172	122.822828	1.1.1.1	2.2.2.2	ISAKMP	434	Identity Protection (Main Mode)
173	122.839759	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
174	123.333749	2.2.2.2	1.1.1.1	ISAKMP	742	Identity Protection (Main Mode)
175	123.343473	1.1.1.1	2.2.2.2	ISAKMP	742	Identity Protection (Main Mode)
177	124.352601	1.1.1.1	2.2.2.2	ISAKMP	434	Identity Protection (Main Mode)
178	124.364442	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
179	124.859609	2.2.2.2	1.1.1.1	ISAKMP	742	Identity Protection (Main Mode)
180	124.870661	1.1.1.1	2.2.2.2	ISAKMP	742	Identity Protection (Main Mode)
182	125.872345	1.1.1.1	2.2.2.2	ISAKMP	434	Identity Protection (Main Mode)
183	125.899941	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
184	126.383736	2.2.2.2	1.1.1.1	ISAKMP	742	Identity Protection (Main Mode)
185	126.403320	1.1.1.1	2.2.2.2	ISAKMP	742	Identity Protection (Main Mode)
187	127.400181	1.1.1.1	2.2.2.2	ISAKMP	434	Identity Protection (Main Mode)
188	127.422952	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
189	127.908572	2.2.2.2	1.1.1.1	ISAKMP	742	Identity Protection (Main Mode)
190	127.926655	1.1.1.1	2.2.2.2	ISAKMP	742	Identity Protection (Main Mode)
192	128.921906	1.1.1.1	2.2.2.2	ISAKMP	434	Identity Protection (Main Mode)
193	128.946909	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
194	129.431559	2.2.2.2	1.1.1.1	ISAKMP	742	Identity Protection (Main Mode)
195	129.452803	1.1.1.1	2.2.2.2	ISAKMP	742	Identity Protection (Main Mode)
196	129.969676	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
197	130.445735	1.1.1.1	2.2.2.2	ISAKMP	434	Identity Protection (Main Mode)
198	130.482267	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
203	140.463165	1.1.1.1	2.2.2.2	ISAKMP	434	Identity Protection (Main Mode)
204	140.508341	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
210	150.467697	1.1.1.1	2.2.2.2	ISAKMP	434	Identity Protection (Main Mode)
211	150.510451	2.2.2.2	1.1.1.1	ISAKMP	434	Identity Protection (Main Mode)
212	151.425759	2.2.2.2	1.1.1.1	ISAKMP	210	Identity Protection (Main Mode)
213	151.428092	1.1.1.1	2.2.2.2	ISAKMP	150	Identity Protection (Main Mode)
214	151.445928	2.2.2.2	1.1.1.1	ISAKMP	414	Identity Protection (Main Mode)
215	151.458714	1.1.1.1	2.2.2.2	ISAKMP	434	Identity Protection (Main Mode)
216	151.496294	2.2.2.2	1.1.1.1	ISAKMP	742	Identity Protection (Main Mode)
217	151.729487	1.1.1.1	2.2.2.2	ISAKMP	210	Identity Protection (Main Mode)
218	151.747897	2.2.2.2	1.1.1.1	ISAKMP	150	Identity Protection (Main Mode)

Figure 19: Repeated ISAKMP negotiation attempts after certificate revocation

The repeated ISAKMP packets indicate that the routers continuously attempt to establish an IKEv1 Security Association but fail during the authentication phase.

This behavior was confirmed using IOS verification commands.

```
R1#sh crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst src state conn-id status
1.1.1.1 2.2.2.2 MM_KEY_EXCH 1040 ACTIVE
1.1.1.1 2.2.2.2 MM_KEY_EXCH 1038 ACTIVE
1.1.1.1 2.2.2.2 MM_NO_STATE 1036 ACTIVE (deleted)
1.1.1.1 2.2.2.2 MM_NO_STATE 1034 ACTIVE (deleted)
2.2.2.2 1.1.1.1 MM_KEY_EXCH 1041 ACTIVE
2.2.2.2 1.1.1.1 MM_NO_STATE 1039 ACTIVE (deleted)
2.2.2.2 1.1.1.1 MM_NO_STATE 1037 ACTIVE (deleted)
```

```
IPv6 Crypto ISAKMP SA
```

The ISAKMP state remains in `MM_KEY_EXCH`, indicating that Main Mode negotiation does not complete successfully.

A successfully established tunnel would normally reach the `QM_IDLE` state.

3.9.3 Absence of IPsec Security Associations

The IPsec SA database confirms that no operational IPsec tunnel was created.

```
R1#sh crypto ipsec sa
```

```
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 1.1.1.1

  protected vrf: (none)
  local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  current_peer 2.2.2.2 port 500
    PERMIT, flags={origin_is_acl,} #pkts encaps: 0,
    #pkts encrypt: 0, #pkts digest: 0 #pkts decaps: 0,
    #pkts decrypt: 0, #pkts verify: 0 #pkts compressed: 0,
    #pkts decompressed: 0 #pkts not compressed: 0,
    #pkts compr. failed: 0 #pkts not decompressed: 0,
    #pkts decompress failed: 0
    #send errors 0, #recv errors 0

  local crypto endpt.: 1.1.1.1, remote crypto endpt.: 2.2.2.2
  plaintext mtu 1500, path mtu 1500, ip mtu 1500, ip mtu idb
GigabitEthernet0/0
  current outbound spi: 0x0(0)
  PFS (Y/N): N, DH group: none

  inbound esp sas:

  inbound ah sas:

  inbound pcp sas:

  outbound esp sas:

  outbound ah sas:

  outbound pcp sas:
```

```
R1#
```

The counters remain at zero because IPsec encryption was never activated. Additionally, no ESP Security Associations were present:

inbound esp sas:
outbound esp sas:

This confirms that Phase 2 IPsec negotiation could not be completed.

3.9.4 Explanation of Communication Failure

Communication fails because the certificates used by R1 and R2 were revoked by the Certification Authority.

After reboot, the routers verify certificate validity during IKEv1 authentication. The CA indicates that the peer certificates are no longer trusted because they are listed in the Certificate Revocation List (CRL).

As a result:

- RSA signature authentication fails
- IKEv1 Main Mode negotiation cannot complete
- ISAKMP Security Associations are not successfully established
- IPsec Security Associations are never created
- ESP encrypted traffic cannot be generated
- ICMP communication between PC1 and PC2 fails

The experiment demonstrates the importance of certificate revocation in PKI-based IPsec infrastructures, ensuring that compromised or invalid certificates cannot be used to establish trusted VPN connections.

4 IPsec with IKEv2

4.1 Introduction

This section describes the configuration and analysis of an IPsec VPN using IKEv2 authentication with pre-shared keys. The objective is to establish a protected tunnel between routers R1 and R2 through the intermediate router RA and analyze the establishment and operation of IKEv2 and IPsec Security Associations (SAs).

4.2 IKEv2 and IPsec Configuration Verification

The IKEv2 and IPsec configuration was verified on both R1 and R2 using IOS verification commands.

The following elements were validated:

- IKEv2 proposals defining AES-CBC-256 encryption, SHA512 integrity, SHA512 PRF, and Diffie-Hellman Group 16;

- IKEv2 policies associated with the configured proposals;
- IPSec transform sets and profiles used for tunnel protection;
- active IPSec Security Associations and encapsulation parameters.

Figures 20, 21, 22, and 23 present the verification outputs collected from both routers.

```
R1#show crypto ikev2 proposal
IKEv2 proposal: default
  Encryption : AES-CBC-256 AES-CBC-192 AES-CBC-128
  Integrity   : SHA512 SHA384 SHA256 SHA96 MD596
  PRF         : SHA512 SHA384 SHA256 SHA1 MD5
  DH Group    : DH_GROUP_1536_MODP/Group 5 DH_GROUP_1024_MODP/Group 2
IKEv2 proposal: myProposal
  Encryption : AES-CBC-256
  Integrity   : SHA512
  PRF         : SHA512
  DH Group    : DH_GROUP_4096_MODP/Group 16
R1#
```

R1

```
R2#show crypto ikev2 proposal
IKEv2 proposal: default
  Encryption : AES-CBC-256 AES-CBC-192 AES-CBC-128
  Integrity   : SHA512 SHA384 SHA256 SHA96 MD596
  PRF         : SHA512 SHA384 SHA256 SHA1 MD5
  DH Group    : DH_GROUP_1536_MODP/Group 5 DH_GROUP_1024_MODP/Group 2
IKEv2 proposal: myProposal
  Encryption : AES-CBC-256
  Integrity   : SHA512
  PRF         : SHA512
  DH Group    : DH_GROUP_4096_MODP/Group 16
R2#
```

R2

Figure 20: IKEv2 proposal verification

```
R1#sh crypto ikev2 policy

IKEv2 policy : default
  Match fvrfl : any
  Match address local : any
  Proposal    : default

IKEv2 policy : myIKEv2Policy
  Match fvrfl : global
  Match address local : any
  Proposal    : myProposal
R1#
```

R1

```
R2#sh crypto ikev2 policy

IKEv2 policy : default
  Match fvrfl : any
  Match address local : any
  Proposal    : default

IKEv2 policy : myIKEv2Policy
  Match fvrfl : global
  Match address local : any
  Proposal    : myProposal
R2#
```

R2

Figure 21: IKEv2 policy verification

```
R1#show crypto ipsec profile
IPSEC profile default
  Security association lifetime: 4608000 kilobytes/3600 seconds
  Responder-Only (Y/N): N
  PFS (Y/N): N
  Mixed-mode : Disabled
  Transform sets={
    default: { esp-aes esp-sha-hmac },
  }

IPSEC profile myIPSecProfile
  IKEv2 Profile: myIKEv2Profile
  Security association lifetime: 4608000 kilobytes/3600 seconds
  Responder-Only (Y/N): N
  PFS (Y/N): N
  Mixed-mode : Disabled
  Transform sets={
    myTSet: { esp-aes esp-sha-hmac },
  }
R1#
```

R1

```
R2#show crypto ipsec profile
IPSEC profile default
  Security association lifetime: 4608000 kilobytes/3600 seconds
  Responder-Only (Y/N): N
  PFS (Y/N): N
  Mixed-mode : Disabled
  Transform sets={
    default: { esp-aes esp-sha-hmac },
  }

IPSEC profile myIPSecProfile
  IKEv2 Profile: myIKEv2Profile
  Security association lifetime: 4608000 kilobytes/3600 seconds
  Responder-Only (Y/N): N
  PFS (Y/N): N
  Mixed-mode : Disabled
  Transform sets={
    myTSet: { esp-aes esp-sha-hmac },
  }
R2#
```

R2

Figure 22: IPSec profile verification

```

R1#show crypto ipsec sa
Interface: Tunnel0
Crypto map tag: Tunnel0-head-0, local addr 1.1.1.1

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 2.2.2.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 41, #pkts encrypt: 41, #pkts digest: 41
  #pkts decaps: 40, #pkts decrypt: 40, #pkts verify: 40
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0

local crypto endpt.: 1.1.1.1, remote crypto endpt.: 2.2.2.2
plaintext mtu 1438, path mtu 1500, ip mtu 1500, ip mtu idb GIGabitEthernet0/0
current outbound spi: 0xB112069E(2970748574)
PFS (Y/N): N, DH group: none

inbound esp sas:
  spi: 0x5CF1C655(1559348821)
    transform: esp-aes esp-sha-hmac ,
    in use settings = (Tunnel, )
    conn id: 1, flow_id: SW:1, sibling_flags 80000040, crypto map: Tunnel0-head-0
  sa timing: remaining key lifetime (k/sec): (4333709/3289)
  IV size: 16 bytes
  replay detection support: Y
  Status: ACTIVE(ACTIVE)

inbound ah sas:

inbound pcp sas:

outbound esp sas:
  spi: 0xB112069E(2970748574)
    transform: esp-aes esp-sha-hmac ,
    in use settings = (Tunnel, )
    conn id: 2, flow_id: SW:2, sibling_flags 80000040, crypto map: Tunnel0-head-0
  sa timing: remaining key lifetime (k/sec): (4333709/3289)
  IV size: 16 bytes
  replay detection support: Y
  Status: ACTIVE(ACTIVE)

outbound ah sas:

outbound pcp sas:

```

```

R2#show crypto ipsec sa
Interface: Tunnel0
Crypto map tag: Tunnel0-head-0, local addr 2.2.2.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
current_peer 1.1.1.1 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 39, #pkts encrypt: 39, #pkts digest: 39
  #pkts decaps: 38, #pkts decrypt: 38, #pkts verify: 38
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0

local crypto endpt.: 2.2.2.2, remote crypto endpt.: 1.1.1.1
plaintext mtu 1438, path mtu 1500, ip mtu 1500, ip mtu idb GIGabitEthernet0/0
current outbound spi: 0x5CF1C655(1559348821)
PFS (Y/N): N, DH group: none

inbound esp sas:
  spi: 0xB112069E(2970748574)
    transform: esp-aes esp-sha-hmac ,
    in use settings = (Tunnel, )
    conn id: 2, flow_id: SW:2, sibling_flags 80000040, crypto map: Tunnel0-head-0
  sa timing: remaining key lifetime (k/sec): (4219799/3307)
  IV size: 16 bytes
  replay detection support: Y
  Status: ACTIVE(ACTIVE)

inbound ah sas:

inbound pcp sas:

outbound esp sas:
  spi: 0x5CF1C655(1559348821)
    transform: esp-aes esp-sha-hmac ,
    in use settings = (Tunnel, )
    conn id: 1, flow_id: SW:1, sibling_flags 80000040, crypto map: Tunnel0-head-0
  sa timing: remaining key lifetime (k/sec): (4219799/3307)
  IV size: 16 bytes
  replay detection support: Y
  Status: ACTIVE(ACTIVE)

outbound ah sas:

outbound pcp sas:

```

R1

R2

Figure 23: Active IPsec Security Associations

The verification outputs confirm that the tunnel was successfully established and that encrypted traffic was being encapsulated and decapsulated through ESP Security Associations.

4.3 Tunnel Establishment and Connectivity Test

After configuration, end-to-end connectivity between PC1 and PC2 was verified using ICMP echo requests.

Successful communication confirmed that:

- the IPsec tunnel was operational;
- encrypted traffic was correctly forwarded between endpoints;
- routing across the protected tunnel was functional.

Figure 24 shows successful ICMP communication between the end hosts through the IPsec tunnel.

145	142.256895	200.1.1.1	224.0.0.5	OSPF	94 Hello Packet
146	142.412693	200.1.1.10	224.0.0.5	OSPF	94 Hello Packet
147	144.213575	1.1.1.1	2.2.2.2	ESP	182 ESP (SPI=0x21302c02)
148	144.238025	2.2.2.2	1.1.1.1	ESP	182 ESP (SPI=0x882526f5)
149	144.253833	1.1.1.1	2.2.2.2	ESP	182 ESP (SPI=0x21302c02)
150	144.278453	2.2.2.2	1.1.1.1	ESP	182 ESP (SPI=0x882526f5)
151	144.294654	1.1.1.1	2.2.2.2	ESP	182 ESP (SPI=0x21302c02)
152	144.318756	2.2.2.2	1.1.1.1	ESP	182 ESP (SPI=0x882526f5)
153	145.477142	0c:35:d9:73:00:00	0c:35:d9:73:00:00	LOOP	60 Reply
154	146.080061	ca:01:22:46:00:00	ca:01:22:46:00:00	LOOP	60 Reply
155	150.752928	1.1.1.1	2.2.2.2	ESP	182 ESP (SPI=0x21302c02)
156	150.793220	2.2.2.2	1.1.1.1	ESP	182 ESP (SPI=0x882526f5)
157	150.813930	1.1.1.1	2.2.2.2	ESP	182 ESP (SPI=0x21302c02)
158	150.853670	2.2.2.2	1.1.1.1	ESP	182 ESP (SPI=0x882526f5)
159	150.874735	1.1.1.1	2.2.2.2	ESP	182 ESP (SPI=0x21302c02)
160	150.914462	2.2.2.2	1.1.1.1	ESP	182 ESP (SPI=0x882526f5)
161	150.935790	1.1.1.1	2.2.2.2	ESP	182 ESP (SPI=0x21302c02)
162	150.975803	2.2.2.2	1.1.1.1	ESP	182 ESP (SPI=0x882526f5)
163	150.996671	1.1.1.1	2.2.2.2	ESP	182 ESP (SPI=0x21302c02)
164	151.036243	2.2.2.2	1.1.1.1	ESP	182 ESP (SPI=0x882526f5)
165	151.421387	200.1.1.1	224.0.0.5	OSPF	94 Hello Packet
166	152.018463	2.2.2.2	1.1.1.1	ESP	166 ESP (SPI=0x882526f5)
167	152.068810	200.1.1.10	224.0.0.5	OSPF	94 Hello Packet
168	152.236657	1.1.1.1	2.2.2.2	ESP	166 ESP (SPI=0x21302c02)

»	Frame 147: 182 bytes on wire (1456 bits), 182 bytes captured (1456 bits) on interface -, id 0
»	Ethernet II, Src: 0c:35:d9:73:00:00 (0c:35:d9:73:00:00), Dst: ca:01:22:46:00:00 (ca:01:22:46:00:00)
»	Internet Protocol Version 4, Src: 1.1.1.1, Dst: 2.2.2.2
»	Encapsulating Security Payload
	ESP SPI: 0x21302c02 (556805122)
	ESP Sequence: 14

Figure 24: Successful ICMP connectivity between PC1 and PC2

4.4 Analysis of IKE_SA_INIT Exchange

Traffic captures were obtained on interface G0/0 of R1 during tunnel establishment. The IKE_SA_INIT exchange is responsible for:

- negotiation of cryptographic algorithms;
- exchange of Diffie-Hellman public values;
- generation of shared secret material;
- exchange of nonces;
- establishment of the initial IKE Security Association.

Figure 26 presents the captured IKEv2 exchange messages.

isakmp						
No.	Time	Source	Destination	Protocol	Length	Info
101	122.645881	2.2.2.2	1.1.1.1	ISAKMP	816	IKE_SA_INIT MID=00 Initiator
102	122.784401	1.1.1.1	2.2.2.2	ISAKMP	816	IKE_SA_INIT MID=00 Responder
103	122.945501	2.2.2.2	1.1.1.1	ISAKMP	666	IKE_AUTH MID=01 Initiator
104	122.973294	1.1.1.1	2.2.2.2	ISAKMP	362	IKE_AUTH MID=01 Responder

▶ Frame 101: 816 bytes on wire (6528 bits), 816 bytes captured (6528 bits) on interface -, id 0
 ▶ Ethernet II, Src: ca:01:22:46:00:00 (ca:01:22:46:00:00), Dst: 0c:35:d9:73:00:00 (0c:35:d9:73:00:00)
 ▶ Internet Protocol Version 4, Src: 2.2.2.2, Dst: 1.1.1.1
 ▶ User Datagram Protocol, Src Port: 500, Dst Port: 500
 ▼ Internet Security Association and Key Management Protocol
 Initiator SPI: 6f4bfad45cd2562a
 Responder SPI: 0000000000000000
 Next payload: Security Association (33)
 ▶ Version: 2.0
 Exchange type: IKE_SA_INIT (34)
 ▶ Flags: 0x08 (Initiator, No higher version, Request)
 Message ID: 0x00000000
 Length: 774
 ▼ Payload: Security Association (33)
 Next payload: Key Exchange (34)
 0... .. = Critical Bit: Not critical
 .000 0000 = Reserved: 0x00
 Payload length: 48
 ▶ Payload: Proposal (2) # 1
 ▼ Payload: Key Exchange (34)
 Next payload: Nonce (40)
 0... .. = Critical Bit: Not critical
 .000 0000 = Reserved: 0x00
 Payload length: 520
 DH Group #: 4096 bit MODP group (16)
 Reserved: 0000
 Key Exchange Data: 7c12973a5156c65e69aea99705452c72c4e969da6e0c44df2b2891bc8606783496cc2f5c...
 ▼ Payload: Nonce (40)
 Next payload: Vendor ID (43)
 0... .. = Critical Bit: Not critical
 .000 0000 = Reserved: 0x00
 Payload length: 36
 Nonce DATA: 3693f79f52a52608e88b19133907eb22b221f0e57a6ad045b644eb96130ad870
 ▶ Payload: Vendor ID (43) : Cisco Delete Reason Supported
 ▶ Payload: Vendor ID (43) : Cisco VPN Revision 2
 ▶ Payload: Vendor ID (43) : Cisco Dynamic Route Supported
 ▶ Payload: Vendor ID (43) : Cisco FlexVPN Supported
 ▶ Payload: Notify (41) - NAT_DETECTION_SOURCE_IP
 ▶ Payload: Notify (41) - NAT_DETECTION_DESTINATION_IP

Figure 25: Contents of the IKE_SA_INIT exchange

```

Payload length: 40
▼ Payload: Proposal (2) # 1
  Next payload: NONE / No Next Payload (0)
  Reserved: 00
  Payload length: 44
  Proposal number: 1
  Protocol ID: IKE (1)
  SPI Size: 0
  Proposal transforms: 4
  ▼ Payload: Transform (3)
    Next payload: Transform (3)
    Reserved: 00
    Payload length: 12
    Transform Type: Encryption Algorithm (ENCR) (1)
    Reserved: 00
    Transform ID (ENCR): ENCR_AES_CBC (12)
    ▶ Transform Attribute (t=14,l=2): Key Length: 256
  ▼ Payload: Transform (3)
    Next payload: Transform (3)
    Reserved: 00
    Payload length: 8
    Transform Type: Pseudo-random Function (PRF) (2)
    Reserved: 00
    Transform ID (PRF): PRF_HMAC_SHA2_512 (7)
  ▼ Payload: Transform (3)
    Next payload: Transform (3)
    Reserved: 00
    Payload length: 8
    Transform Type: Integrity Algorithm (INTEG) (3)
    Reserved: 00
    Transform ID (INTEG): AUTH_HMAC_SHA2_512_256 (14)
  ▼ Payload: Transform (3)
    Next payload: NONE / No Next Payload (0)
    Reserved: 00
    Payload length: 8
    Transform Type: Diffie-Hellman Group (D-H) (4)
    Reserved: 00
    Transform ID (D-H): 4096 bit MODP group (16)
Payload: Key Exchange (34)

```

Figure 26: Content of the payload: Proposal in the IKE_SA_INIT packet

The capture shows the Security Parameter Indexes (SPIs), Security Association proposals, Diffie-Hellman key exchange payloads, and nonce payloads exchanged between both peers.

The negotiated proposal includes:

- AES-CBC-256 encryption;
- SHA2-512 integrity protection;
- SHA2-512 pseudo-random function (PRF);
- Diffie-Hellman Group 16.

The nonce payloads contribute entropy to the key generation process and ensure freshness of the exchange, protecting against replay attacks.

4.5 SA Deletion and Re-establishment

The command `clear crypto session` was executed to remove the active Security Associations and force a new negotiation process.

Wireshark captures show the deletion of the existing SAs followed by the establishment of new IKEv2 and IPSec Security Associations.

After the previous SAs were removed, new `IKE_SA_INIT` and `IKE_AUTH` exchanges were observed, resulting in the creation of new SPIs and new IPSec Security Associations.

The results confirm that IPSec tunnel establishment using IKEv2 was successfully achieved between R1 and R2.

The analysis of Wireshark captures demonstrated:

- negotiation of cryptographic algorithms during `IKE_SA_INIT`;
- exchange of Diffie-Hellman values and nonces;
- encrypted peer authentication during `IKE_AUTH`;
- successful creation of IPSec Security Associations;
- successful deletion and recreation of SAs after clearing the crypto session.

Cisco IOS verification commands further confirmed the operational state of the tunnel, the establishment of OSPF adjacency through Tunnel0, and the correct operation of encrypted ESP traffic.

4.6 IKEv2 Security Association Rekeying

To analyze the IKEv2 rekeying process, the lifetime of the IKEv2 Security Association was reduced to 180 seconds on R2 using the following configuration:

```
crypto ikev2 profile myIKEv2Profile
  lifetime 180
```

After applying the configuration, the existing Security Associations were cleared using the `clear crypto session` command, allowing a new IKEv2 SA to be established with the updated lifetime.

Wireshark captures were obtained during the rekeying process. Figure 27 presents the exchanged messages associated with IKEv2 rekeying.

318	398.049206	2.2.2.2	1.1.1.1	ISAKMP	762 CREATE_CHILD_SA MID=02 Initiator Request
319	398.208250	1.1.1.1	2.2.2.2	ISAKMP	762 CREATE_CHILD_SA MID=02 Responder Response
320	398.365883	2.2.2.2	1.1.1.1	ISAKMP	138 INFORMATIONAL MID=03 Initiator Request
321	398.380517	1.1.1.1	2.2.2.2	ISAKMP	138 INFORMATIONAL MID=03 Responder Response
322	400.499255	200.2.2.2	224.0.0.5	OSPF	94 Hello Packet
323	400.548177	0c:14:9a:81:00:00	0c:14:9a:81:00:00	LOOP	60 Reply
324	400.606241	200.2.2.10	224.0.0.5	OSPF	94 Hello Packet

```

▶ Frame 318: 762 bytes on wire (6096 bits), 762 bytes captured (6096 bits) on interface -, id 0
▶ Ethernet II, Src: 0c:14:9a:81:00:00 (0c:14:9a:81:00:00), Dst: ca:01:22:46:00:1c (ca:01:22:46:00:1c)
▶ Internet Protocol Version 4, Src: 2.2.2.2, Dst: 1.1.1.1
▶ User Datagram Protocol, Src Port: 500, Dst Port: 500
▼ Internet Security Association and Key Management Protocol
  Initiator SPI: c706886ac926b32f
  Responder SPI: 7328fbe7b3aa94c6
  Next payload: Encrypted and Authenticated (46)
▼ Version: 2.0
  0010 .... = MjVer: 0x2
  .... 0000 = MnVer: 0x0
  Exchange type: CREATE_CHILD_SA (36)
▶ Flags: 0x08 (Initiator, No higher version, Request)
  Message ID: 0x00000002
  Length: 720
▼ Payload: Encrypted and Authenticated (46)
  Next payload: Security Association (33)
  0... .... = Critical Bit: Not critical
  .000 0000 = Reserved: 0x00
  Payload length: 692
  Initialization Vector: 356844f0
  Encrypted Data

```

Figure 27: IKEv2 rekeying messages captured in Wireshark

The captures show that the rekeying procedure does not restart the tunnel establishment from scratch. No new `IKE_SA_INIT` exchange was observed during rekeying. Instead, the process occurs through protected exchanges within the existing secure channel.

The `CREATE_CHILD_SA` exchange is responsible for generating fresh key material and establishing a new Security Association while the previous one is still active. This allows rekeying to occur without interrupting tunnel connectivity.

The evolution of the IKEv2 Security Association was verified using the `show crypto ikev2 sa detailed` command. Figure 28 shows the Security Association state before and after rekeying.

```

R2#show crypto ikev2 sa detailed
IPv4 Crypto IKEv2 SA

Tunnel-id Local Remote fvrf/ivrf Status
1 2.2.2.2/500 1.1.1.1/500 none/none READY
Encr: AES-CBC, keysize: 256, PRF: SHA512, Hash: SHA512, DH Grp:16, Auth sign: PSK, Auth ve
Life/Active Time: 180/69 sec
CE id: 1003, Session-id: 2
Status Description: Negotiation done
Local spi: C706886AC926B32F Remote spi: 7328FBE7B3AA94C6
Local id: 2.2.2.2
Remote id: 1.1.1.1
Local req msg id: 2 Remote req msg id: 0
Local next msg id: 2 Remote next msg id: 0
Local req queued: 2 Remote req queued: 0
Local window: 5 Remote window: 5
DPD configured for 0 seconds, retry 0
Fragmentation not configured.
Dynamic Route Update: disabled
Extended Authentication not configured.
NAT-T is not detected
Cisco Trust Security SGT is disabled
Initiator of SA : Yes

IPv6 Crypto IKEv2 SA

R2#show crypto ikev2 sa detailed
IPv4 Crypto IKEv2 SA

Tunnel-id Local Remote fvrf/ivrf Status
6 2.2.2.2/500 1.1.1.1/500 none/none READY
Encr: AES-CBC, keysize: 256, PRF: SHA512, Hash: SHA512, DH Grp:16, Auth sign: PSK, Auth ve
Life/Active Time: 180/89 sec
CE id: 0, Session-id: 2
Status Description: Negotiation done
Local spi: E18E02B6B2633345 Remote spi: 96BE8FEFC1B8A5F0
Local id: 2.2.2.2
Remote id: 1.1.1.1
Local req msg id: 0 Remote req msg id: 0
Local next msg id: 0 Remote next msg id: 0
Local req queued: 0 Remote req queued: 0
Local window: 5 Remote window: 5
DPD configured for 0 seconds, retry 0
Fragmentation not configured.
Dynamic Route Update: disabled
Extended Authentication not configured.
NAT-T is not detected
Cisco Trust Security SGT is disabled
Initiator of SA : Yes

IPv6 Crypto IKEv2 SA

R2#

```

Figure 28: Evolution of the IKEv2 Security Association during rekeying

The outputs show that rekeying generated new local and remote SPIs while preserving the negotiated cryptographic algorithms and parameters. Additionally, the SA lifetime was refreshed after rekeying, confirming the successful establishment of a new IKEv2 Security Association.

The change in Tunnel-id, from 1 to 6, and SPI values demonstrates that the previous Security Association was replaced by a newly negotiated association.

4.7 IPsec Security Association Rekeying

To analyze the IPsec Security Association rekeying process, the IPsec SA lifetime was reduced to 120 seconds on R1 using the following configuration:

```
crypto ipsec profile myIPsecProfile
set security-association lifetime seconds 120
```

After saving the configuration, router R1 was restarted to ensure that the updated lifetime was applied to newly established IPsec Security Associations.

Wireshark captures were obtained during the rekeying process. Figure 29 presents the exchanged messages associated with IPsec SA rekeying.

186	206.938635	ca:01:22:46:00:00	ca:01:22:46:00:00	LOOP	60 Reply
187	206.979767	1.1.1.1	2.2.2.2	ESP	166 ESP (SPI=0xe9b50866)
188	207.390333	1.1.1.1	2.2.2.2	ISAKMP	266 CREATE_CHILD_SA MID=02 Initiator Request
189	207.401480	2.2.2.2	1.1.1.1	ISAKMP	266 CREATE_CHILD_SA MID=02 Responder Response
190	207.412676	1.1.1.1	2.2.2.2	ISAKMP	138 INFORMATIONAL MID=03 Initiator Request
191	207.431676	2.2.2.2	1.1.1.1	ISAKMP	138 INFORMATIONAL MID=03 Responder Response
192	209.092270	200.1.1.10	224.0.0.5	OSPF	94 Hello Packet
193	210.512807	2.2.2.2	1.1.1.1	ESP	166 ESP (SPI=0x951074b7)
194	211.131286	200.1.1.1	224.0.0.5	OSPF	94 Hello Packet
195	216.113295	1.1.1.1	2.2.2.2	ESP	166 ESP (SPI=0x59ef318e)
196	216.901307	0c:35:d9:73:00:00	0c:35:d9:73:00:00	LOOP	60 Reply
197	216.931805	ca:01:22:46:00:00	ca:01:22:46:00:00	LOOP	60 Reply
198	218.163844	200.1.1.10	224.0.0.5	OSPF	94 Hello Packet

- Frame 188: 266 bytes on wire (2128 bits), 266 bytes captured (2128 bits) on interface -, id 0 - Ethernet II, Src: 0c:35:d9:73:00:00 (0c:35:d9:73:00:00), Dst: ca:01:22:46:00:00 (ca:01:22:46:00:00) - Internet Protocol Version 4, Src: 1.1.1.1, Dst: 2.2.2.2 - User Datagram Protocol, Src Port: 500, Dst Port: 500 - Internet Security Association and Key Management Protocol - Initiator SPI: 6abadc0329dcdca4 - Responder SPI: 15504939108ed5ed - Next payload: Encrypted and Authenticated (46) - Version: 2.0 0010 = MjVer: 0x2 0000 = MnVer: 0x0 - Exchange type: CREATE_CHILD_SA (36) - Flags: 0x08 (Initiator, No higher version, Request) 1... = Initiator: Initiator ...0 = Version: No higher version ..0. = Response: Request
- Message ID: 0x00000002 - Length: 224 - Payload: Encrypted and Authenticated (46) - Next payload: Notify (41) 0... = Critical Bit: Not critical .000 0000 = Reserved: 0x00 - Payload length: 196 - Initialization Vector: 334b883b - Encrypted Data

Figure 29: IPsec Security Association rekeying captured in Wireshark

Although both IKEv2 SA rekeying and IPsec SA rekeying use CREATE_CHILD_SA exchanges, the affected Security Associations differ. During IKEv2 SA rekeying, the IKE Security Association itself is replaced and new IKE SPIs are generated. During IPsec SA rekeying, only the ESP CHILD_SA is replaced while the IKEv2 Security Association remains active.

The evolution of the IPsec Security Associations was verified using the `show crypto ipsec sa` command. Figure 30 presents the Security Association state before and after rekeying.

```

R1#show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 1.1.1.1

  protected vrf: (none)
  local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  current_peer 2.2.2.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 12, #pkts encrypt: 12, #pkts digest: 12
    #pkts decaps: 11, #pkts decrypt: 11, #pkts verify: 11
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

  local crypto endpt.: 1.1.1.1, remote crypto endpt.: 2.2.2.2
  plaintext mtu 1438, path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/0
  current outbound spi: 0xE9B50860(3920955494)
  PFS (Y/N): N, DH group: none

  inbound esp sas:
    spi: 0xEE230284(3995271812)
    transform: esp-aes esp-sha-hmac ,
    in use settings ={Tunnel,}
    conn id: 2, flow_id: SW:2, sibling_flags 80000040, crypto map: Tunnel0-head-0
    sa timing: remaining key lifetime (k/sec): (4147775/89)
    IV size: 16 bytes
    replay detection support: Y
    Status: ACTIVE(ACTIVE)

  inbound ah sas:

  inbound pcp sas:

  outbound esp sas:
    spi: 0xE9B50860(3920955494)
    transform: esp-aes esp-sha-hmac ,
    in use settings ={Tunnel,}
    conn id: 1, flow_id: SW:1, sibling_flags 80000040, crypto map: Tunnel0-head-0
    sa timing: remaining key lifetime (k/sec): (4147775/89)
    IV size: 16 bytes
    replay detection support: Y
    Status: ACTIVE(ACTIVE)

  outbound ah sas:

  outbound pcp sas:

```

Before rekeying

```

R1#show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 1.1.1.1

  protected vrf: (none)
  local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  current_peer 2.2.2.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 18, #pkts encrypt: 18, #pkts digest: 18
    #pkts decaps: 17, #pkts decrypt: 17, #pkts verify: 17
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

  local crypto endpt.: 1.1.1.1, remote crypto endpt.: 2.2.2.2
  plaintext mtu 1438, path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/0
  current outbound spi: 0x59EF318E(1508848014)
  PFS (Y/N): N, DH group: none

  inbound esp sas:
    spi: 0x951074B7(2500883639)
    transform: esp-aes esp-sha-hmac ,
    in use settings ={Tunnel,}
    conn id: 6, flow_id: SW:6, sibling_flags 80000040, crypto map: Tunnel0-head-0
    sa timing: remaining key lifetime (k/sec): (4261406/112)
    IV size: 16 bytes
    replay detection support: Y
    Status: ACTIVE(ACTIVE)

  inbound ah sas:

  inbound pcp sas:

  outbound esp sas:
    spi: 0x59EF318E(1508848014)
    transform: esp-aes esp-sha-hmac ,
    in use settings ={Tunnel,}
    conn id: 5, flow_id: SW:5, sibling_flags 80000040, crypto map: Tunnel0-head-0
    sa timing: remaining key lifetime (k/sec): (4261407/112)
    IV size: 16 bytes
    replay detection support: Y
    Status: ACTIVE(ACTIVE)

  outbound ah sas:

  outbound pcp sas:

```

After rekeying

Figure 30: Evolution of the IPSec Security Associations during rekeying

The outputs show that the inbound and outbound ESP SPIs changed after rekeying, confirming that new IPSec Security Associations were created. Additionally, the Security Association lifetime was refreshed after rekeying while the negotiated cryptographic parameters remained unchanged.

The packet counters continued increasing throughout the process, demonstrating that encrypted traffic continued uninterrupted during the IPSec SA rekeying operation.

Unlike IKEv2 Security Association rekeying, which replaces the IKE SA itself, IPSec rekeying only replaces the CHILD_SA used for ESP traffic protection while preserving the already established IKEv2 Security Association.

5 DMVPN Phase 3

The objective of this experiment was to configure and analyze a DMVPN Phase 3 network using RIP, NHRP redirects, and NHRP shortcuts. The behavior of DMVPN Phase 3 was compared with DMVPN Phase 1 and Phase 2 using Cisco IOS commands and Wireshark captures.

5.1 Topology

Figure 31 shows the network topology used in the experiment. Router R1 operated as the DMVPN hub, while R2 and R3 operated as spokes.

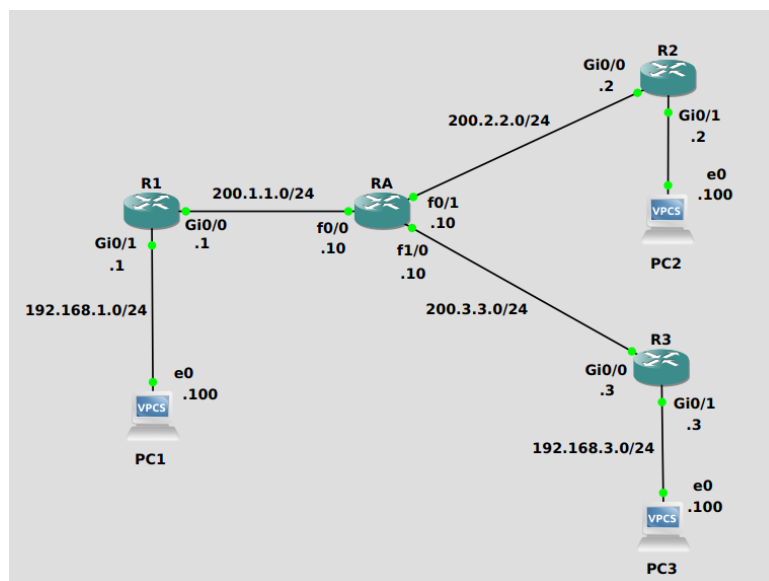


Figure 31: DMVPN Phase 3 topology

5.2 DMVPN Phase 3 Configuration

DMVPN Phase 3 was configured using:

- mGRE tunnels;
- RIP version 2 in the overlay network;
- NHRP redirects on the hub;
- NHRP shortcuts on the spokes.

5.3 Role of `ip summary-address rip`

The command made the hub advertise a default route to all spokes. As a result, the initial traffic was forwarded through the hub before a direct spoke-to-spoke tunnel was created.

Figure 32 shows the RIP default route learned by R2.

```

R2#sh ip route rip
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 10.10.10.1 to network 0.0.0.0

R*    0.0.0.0/0 [120/1] via 10.10.10.1, 00:00:12, Tunnel0
R2#sh ip nhrp
10.10.10.1/32 via 10.10.10.1
      Tunnel0 created 00:06:45, never expire
      Type: static, Flags:
      NBMA address: 1.1.1.1

```

Figure 32: Default route learned through RIP

5.4 NHRP Redirects and Shortcuts

Initially, the spokes only knew the NBMA address of the hub. After the first packets were forwarded through the hub, R1 generated an NHRP redirect informing the spoke that a better path existed.

R2 then sent an NHRP Resolution Request and dynamically learned the NBMA address of R3. This created a direct spoke-to-spoke shortcut.

Figure 33 shows the DMVPN dynamic shortcut entry learned by R2.

```

R2#sh dmvpn
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
       N - NATed, L - Local, X - No Socket
       T1 - Route Installed, T2 - Nexthop-override
       C - CTS Capable, I2 - Temporary
       # Ent --> Number of NHRP entries with same NBMA peer
       NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
       UpDn Time --> Up or Down Time for a Tunnel
=====

Interface: Tunnel0, IPv4 NHRP Details
Type:Spoke, NHRP Peers:2,

# Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
-----
  1 1.1.1.1          10.10.10.1    UP 00:08:35    S
  2 3.3.3.3          10.10.10.3    UP 00:07:57   DT1
          10.10.10.3    UP 00:07:57   DT1

```

Figure 33: Dynamic DMVPN shortcut entry

5.5 Traffic Analysis

Figure 34 shows the traceroute between PC2 and PC3. The traffic was forwarded directly between spokes after the NHRP shortcut was created.

```
PC2> trace 192.168.3.100
trace to 192.168.3.100, 8 hops max, press Ctrl+C to stop
 1  192.168.2.2    0.651 ms  0.613 ms  0.557 ms
 2  10.10.10.3    14.728 ms  19.898 ms  19.887 ms
 3  *192.168.3.100  19.725 ms (ICMP type:3, code:3, Destination port unreachable)
```

Figure 34: Traceroute between spokes

5.6 Wireshark Analysis

Wireshark captures confirmed the operation of DMVPN Phase 3.

Figure 35 shows the NHRP Traffic Indication packets generated by the hub router. These packets informed the spokes that a better path existed.

169 276.503534	1.1.1.1	2.2.2.2	NHRP	122 NHRP Traffic Indication
172 276.525054	1.1.1.1	3.3.3.3	NHRP	122 NHRP Traffic Indication
173 276.533183	2.2.2.2	1.1.1.1	NHRP	110 NHRP Resolution Request, ID=2
174 276.535825	1.1.1.1	3.3.3.3	NHRP	130 NHRP Resolution Request, ID=2
175 276.553515	3.3.3.3	1.1.1.1	NHRP	110 NHRP Resolution Request, ID=2
176 276.554695	1.1.1.1	2.2.2.2	NHRP	130 NHRP Resolution Request, ID=2

Figure 35: NHRP Traffic Indication packets

Figure 36 shows the NHRP Resolution Request and Reply packets exchanged during the shortcut creation process.

79 92.269870	192.168.2.100	192.168.3.100	ICMP	122 Echo (ping) request id=0xb4ae, seq=1/256, ttl=63 (reply in 82)
80 92.284426	1.1.1.1	2.2.2.2	NHRP	122 NHRP Traffic Indication
81 92.295118	2.2.2.2	1.1.1.1	NHRP	110 NHRP Resolution Request, ID=2
82 92.304876	192.168.3.100	192.168.2.100	ICMP	122 Echo (ping) reply id=0xb4ae, seq=1/256, ttl=62 (request in 79)
83 92.325231	3.3.3.3	2.2.2.2	NHRP	158 NHRP Resolution Reply, ID=2, Code=Success
84 92.335360	1.1.1.1	2.2.2.2	NHRP	130 NHRP Resolution Request, ID=2
85 92.338501	2.2.2.2	3.3.3.3	NHRP	158 NHRP Resolution Reply, ID=2, Code=Success

Figure 36: NHRP Resolution Request and Reply

5.7 Role of RIP Split Horizon

RIP split horizon prevents the hub from advertising routes learned from one spoke back to the same DMVPN interface. This forces the initial spoke-to-spoke traffic to pass through the hub, allowing NHRP redirects and shortcuts to operate correctly.

5.8 Comparison Between DMVPN Phases

- DMVPN Phase 1 uses point-to-point tunnels between the hub and the spokes. All spoke-to-spoke traffic is forwarded through the hub router, which creates a centralized but less scalable solution.
- DMVPN Phase 2 introduces dynamic spoke-to-spoke communication using NHRP. The routing protocol preserves the original next-hop information, allowing spokes to establish direct tunnels after the NHRP resolution process.

- DMVPN Phase 3 improves scalability by using route summarization and NHRP redirects. Initially, traffic is forwarded through the hub router. The hub then sends NHRP redirect messages, allowing the spokes to dynamically create shortcuts and establish direct communication.

5.9 Experimental Validation

The experimental behavior matched the theoretical explanation of DMVPN Phase 3:

- RIP advertised a summarized default route;
- the hub generated NHRP redirects;
- spokes dynamically created shortcuts;
- direct spoke-to-spoke communication was established after the first packets.

5.10 Comparison Between AI Explanation and Experimental Results

The AI explanation stated that RIP is used to advertise summarized routes from the hub to the spokes, while NHRP redirects and shortcuts are responsible for dynamically creating direct spoke-to-spoke communication.

The experimental behavior observed in the laboratory matched this explanation. The RIP routing table on R2 showed that the hub advertised a default route.

Wireshark captures confirmed the operation of NHRP redirects and shortcuts. The hub generated NHRP Traffic Indication packets, followed by NHRP Resolution Request and Resolution Reply packets exchanged between the spokes. After this process, a dynamic shortcut entry appeared in the DMVPN table, allowing direct spoke-to-spoke communication.

These results confirmed the theoretical behavior of DMVPN Phase 3.

6 VRFs and BGP-MPLS L3VPNs

6.1 Introduction

This laboratory exercise explored the implementation of a BGP/MPLS Layer 3 VPN using GNS3 and Cisco IOS routers. Two customer organizations (Blue and Red), each with two geographically separated sites, were interconnected through a provider MPLS backbone composed of PE and P routers. Virtual Routing and Forwarding (VRF) instances were used to isolate customer routing information, while MP-BGP distributed VPN routes and MPLS labels between PE routers. The objective of the exercise was to analyze how MPLS labels, VRFs, and MP-BGP cooperate to provide scalable and isolated Layer 3 VPN connectivity across a shared provider infrastructure.

6.2 Ping Between Sites and Loopback Sourcing

Connectivity between the sites of each organization was verified by executing sourced pings between the CE routers. Figure 37 presents the successful ping from Blue1 to Blue2, and Figure 38 presents the successful ping from Red1 to Red2.

```
B1#ping 192.168.102.1 source 10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.102.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.101.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/56/60 ms
B1#
```

Figure 37: Successful ping from Blue1 to Blue2

```
R1#ping 192.168.112.1 source 10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.112.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.111.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/42/48 ms
R1#
```

Figure 38: Successful ping from Red1 to Red2

The pings are sourced from the loopback interfaces because these are the prefixes advertised into the VPN through the static routes redistributed into MP-BGP. The physical CE interface addresses (e.g. 10.10.1.100) are local to the PE-CE segment and are never injected into the VRF routing table of the remote PE. A ping sourced from the physical interface would reach the destination but the reply would be dropped at the remote PE for lack of a return route. Sourcing from the loopback tests the actual end-to-end VPN path across the advertised prefixes.

6.3 PE1 Routing Tables

The routing architecture of PE1 was analyzed by inspecting the global routing table and the per-VRF routing tables. Figure 39 presents the global table, and Figures 40 and 41 present the VRF tables for the blue and red organizations respectively.

```

PE1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

    1.0.0.0/32 is subnetted, 1 subnets
C      1.1.1.1 is directly connected, Loopback0
    2.0.0.0/32 is subnetted, 1 subnets
O      2.2.2.2 [110/12] via 222.222.1.10, 00:22:13, FastEthernet0/0
    222.222.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      222.222.1.0/24 is directly connected, FastEthernet0/0
L      222.222.1.1/32 is directly connected, FastEthernet0/0
O      222.222.2.0/24 [110/11] via 222.222.1.10, 00:22:13, FastEthernet0/0
PE1#

```

Figure 39

The global routing table contains exclusively provider infrastructure routes: the directly connected provider links (222.222.x.x), the PE and P loopbacks learned via OSPF, and no customer prefixes whatsoever. This reflects the fundamental VRF isolation principle, customer routes exist only within their respective VRF instances and are completely invisible to the global forwarding plane.

```

PE1#sh ip route vrf blue
Routing Table: blue
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      10.10.1.0/24 is directly connected, FastEthernet1/1
L      10.10.1.1/32 is directly connected, FastEthernet1/1
S      192.168.101.0/24 [1/0] via 10.10.1.100
B      192.168.102.0/24 [200/0] via 2.2.2.2, 00:21:26
PE1#

```

Figure 40: Blue vrf routing table in PE1

```

PE1#sh ip route vrf red

Routing Table: red
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       11.11.1.0/24 is directly connected, FastEthernet1/0
L       11.11.1.1/32 is directly connected, FastEthernet1/0
S       192.168.111.0/24 [1/0] via 11.11.1.100
B       192.168.112.0/24 [200/0] via 2.2.2.2, 00:21:32
PE1#

```

Figure 41: Red vrf routing table in PE1

The blue VRF routing table contains three categories of entries. The directly connected entry for 10.10.1.0/24 corresponds to the PE1–Blue1 segment. The static route for 192.168.101.0/24 via 10.10.1.100 was manually configured to reach Blue1’s loopback and is redistributed into MP-BGP. The BGP route for 192.168.102.0/24 via 2.2.2.2 was learned from PE2 through the MP-BGP session, with PE2’s loopback serving as the next hop resolved through the MPLS core. The red VRF table is symmetric, containing the equivalent entries for the red organization’s address space.

The two organizations are isolated from each other because each VRF constitutes an independent routing and forwarding instance. Interfaces are explicitly bound to a single VRF, so incoming packets are looked up exclusively within that VRF’s FIB. Furthermore, route import and export is governed by the Route Target extended community: the blue VRF is configured with `route-target both 1:1` and the red VRF with `route-target both 2:2`, meaning a route advertised by one organization is never imported into the other’s VRF regardless of address overlap.

6.4 MPLS Labels During Ping Analysis

To analyze the MPLS label stack used within the provider network, Wireshark captures were obtained on both the PE1–P and P–PE2 links while executing `ping 192.168.102.1 source 10` from Blue1. Figures 42 and 43 present the captured packets on each segment respectively.

44 37.012223	ca:01:00:00:00:00	ca:01:00:00:00:00	LDP	66 Reply
45 38.509634	192.168.101.1	192.168.102.1	ICMP	122 Echo (ping) request id=0x0004, seq=0/0, ttl=254 (reply in 46)
46 38.542598	192.168.102.1	192.168.101.1	ICMP	118 Echo (ping) reply id=0x0004, seq=0/0, ttl=254 (request in 45)
47 38.570153	192.168.101.1	192.168.102.1	ICMP	122 Echo (ping) request id=0x0004, seq=1/256, ttl=254 (reply in 48)
48 38.597381	192.168.102.1	192.168.101.1	ICMP	118 Echo (ping) reply id=0x0004, seq=1/256, ttl=254 (request in 47)
49 38.613389	222.222.1.10	224.0.0.2	LDP	76 Hello Message
50 38.620555	192.168.101.1	192.168.102.1	ICMP	122 Echo (ping) request id=0x0004, seq=2/512, ttl=254 (reply in 51)
51 38.643835	192.168.102.1	192.168.101.1	ICMP	118 Echo (ping) reply id=0x0004, seq=2/512, ttl=254 (request in 50)
52 38.661976	192.168.101.1	192.168.102.1	ICMP	122 Echo (ping) request id=0x0004, seq=3/768, ttl=254 (reply in 53)
53 38.685427	192.168.102.1	192.168.101.1	ICMP	118 Echo (ping) reply id=0x0004, seq=3/768, ttl=254 (request in 52)
54 38.712385	192.168.101.1	192.168.102.1	ICMP	122 Echo (ping) request id=0x0004, seq=4/1024, ttl=254 (reply in 55)
55 38.735862	192.168.102.1	192.168.101.1	ICMP	118 Echo (ping) reply id=0x0004, seq=4/1024, ttl=254 (request in 54)
56 39.400109	222.222.1.1	224.0.0.5	OSPF	94 Hello Packet
57 40.546537	222.222.1.1	224.0.0.2	LDP	76 Hello Message
58 42.222200	222.222.1.10	224.0.0.2	LDP	76 Hello Message
Frame 45: 122 bytes on wire (976 bits), 122 bytes captured (976 bits) on interface -, id 0				
Ethernet II, Src: ca:01:d8:70:00:00 (ca:01:d8:70:00:00), Dst: c2:03:d9:13:00:00 (c2:03:d9:13:00:00)				
MultiProtocol Label Switching Header, Label: 17, Exp: 0, S: 0, TTL: 254				
0000 0000 0000 0001 0001 = MPLS Label: 17 (0x00011)				
..... = MPLS Experimental Bits: 0				
..... = MPLS Bottom Of Label Stack: 0				
..... 1111 1110 = MPLS TTL: 254				
MultiProtocol Label Switching Header, Label: 19, Exp: 0, S: 1, TTL: 254				
0000 0000 0000 0001 0011 = MPLS Label: 19 (0x00013)				
..... = MPLS Experimental Bits: 0				
..... = MPLS Bottom Of Label Stack: 1				
..... 1111 1101 = MPLS TTL: 254				
Internet Protocol Version 4, Src: 192.168.101.1, Dst: 192.168.102.1				
Internet Control Message Protocol				

Figure 42: Message captured between PE1 and P, displaying 2 labels

8 8.462001	ca:01:00:00:00:00	ca:01:00:00:00:00	LDP	76 Hello Message
9 8.133322	192.168.101.1	192.168.102.1	ICMP	118 Echo (ping) request id=0x0005, seq=0/0, ttl=254 (reply in 10)
10 8.149851	192.168.102.1	192.168.101.1	ICMP	122 Echo (ping) reply id=0x0005, seq=0/0, ttl=254 (request in 9)
11 8.183758	192.168.101.1	192.168.102.1	ICMP	118 Echo (ping) request id=0x0005, seq=1/256, ttl=254 (reply in 12)
12 8.200462	192.168.102.1	192.168.101.1	ICMP	122 Echo (ping) reply id=0x0005, seq=1/256, ttl=254 (request in 11)
13 8.224129	192.168.101.1	192.168.102.1	ICMP	118 Echo (ping) request id=0x0005, seq=2/512, ttl=254 (reply in 14)
14 8.251384	192.168.102.1	192.168.101.1	ICMP	122 Echo (ping) reply id=0x0005, seq=2/512, ttl=254 (request in 13)
15 8.276371	192.168.101.1	192.168.102.1	ICMP	118 Echo (ping) request id=0x0005, seq=3/768, ttl=254 (reply in 16)
16 8.293445	192.168.102.1	192.168.101.1	ICMP	122 Echo (ping) reply id=0x0005, seq=3/768, ttl=254 (request in 15)
17 8.328128	192.168.101.1	192.168.102.1	ICMP	118 Echo (ping) request id=0x0005, seq=4/1024, ttl=254 (reply in 18)
18 8.343817	192.168.102.1	192.168.101.1	ICMP	122 Echo (ping) reply id=0x0005, seq=4/1024, ttl=254 (request in 17)
19 8.937142	222.222.2.2	224.0.0.2	LDP	76 Hello Message
20 9.935069	222.222.2.10	224.0.0.2	LDP	76 Hello Message
21 9.945587	c2:03:d9:13:00:01	c2:03:d9:13:00:01	LDP	66 Reply
Frame 9: 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface -, id 0				
Ethernet II, Src: c2:03:d9:13:00:01 (c2:03:d9:13:00:01), Dst: ca:02:d8:99:00:00 (ca:02:d8:99:00:00)				
MultiProtocol Label Switching Header, Label: 19, Exp: 0, S: 1, TTL: 253				
0000 0000 0000 0001 0011 = MPLS Label: 19 (0x00013)				
..... = MPLS Experimental Bits: 0				
..... = MPLS Bottom Of Label Stack: 1				
..... 1111 1101 = MPLS TTL: 253				
Internet Protocol Version 4, Src: 192.168.101.1, Dst: 192.168.102.1				
Internet Control Message Protocol				

Figure 43: Message captured between P-PE2 displaying only 1 label

On the PE1-P segment, each ICMP packet carries a two-label MPLS stack. The outer label (value 17) is the transport label, distributed by LDP, and is used exclusively to switch the packet across the provider core toward PE2's loopback 2.2.2.2. Its Bottom of Stack (S) bit is set to 0, indicating that additional labels follow beneath it in the stack. The inner label (value 19) is the VPN label, distributed by MP-BGP, and identifies the destination VRF at the egress PE. Its S bit is set to 1, marking it as the last label in the stack, after which the original IP packet begins.

It is important to note that the VPN label value 19 was not assigned by PE1, the ingress router, but rather by PE2, the egress router. In MPLS VPN, each PE assigns labels for the prefixes it advertises and communicates them to the remote PE via MP-BGP Update messages. PE2 assigned label 19 for prefix 192.168.102.0/24 and advertised it to PE1, which then imposes that label on all traffic destined for that prefix. This is confirmed by `show bgp vpnv4 unicast vrf blue 192.168.102.0` on PE1 shown in Figure 44, where the outgoing label field shows the value received from PE2. The ingress PE is therefore always responsible for imposing a label that was assigned and advertised by the egress PE.

```
PE1#show bgp vpnv4 unicast vrf blue 192.168.102.0
BGP routing table entry for 1:1:192.168.102.0/24, version 6
Paths: (1 available, best #1, table blue)
  Not advertised to any peer
  Local
    2.2.2.2 (metric 12) from 2.2.2.2 (2.2.2.2)
      Origin incomplete, metric 0, localpref 100, valid, internal, best
      Extended Community: RT:1:1
      mpls labels in/out nolabel/19
PE1#
```

Figure 44

These label values are directly correlated with the MPLS forwarding table of PE1, shown in Figure 45. The entry for 2.2.2.2/32 shows outgoing label 17 via **FastEthernet0/0**, confirming that PE1 imposes the LDP transport label 17 to reach PE2’s loopback through the P router. The VPN entry for 192.168.102.0/24[V] confirms that traffic destined for the blue VRF prefix is forwarded with the label stack observed in the capture.

```
PE1#sh mpls forwarding-table
Local   Outgoing Prefix          Bytes Label  Outgoing  Next Hop
Label   Label    or Tunnel Id    Switched     interface
17      17       2.2.2.2/32      0            Fa0/0     222.222.1.10
18      Pop Label 222.222.2.0/24  0            Fa0/0     222.222.1.10
19      No Label  192.168.101.0/24[V] \
                                         2280      Fa1/1     10.10.1.100
20      No Label  192.168.111.0/24[V] \
                                         1140      Fa1/0     11.11.1.100
PE1#
```

Figure 45

```
PE2#sh mpls forwarding-table
Local   Outgoing Prefix          Bytes Label  Outgoing  Next Hop
Label   Label    or Tunnel Id    Switched     interface
17      16       1.1.1.1/32      0            Fa0/0     222.222.2.10
18      Pop Label 222.222.1.0/24  0            Fa0/0     222.222.2.10
19      No Label  192.168.102.0/24[V] \
                                         2280      Fa1/0     10.10.2.100
20      No Label  192.168.112.0/24[V] \
                                         1140      Fa1/1     11.11.2.100
PE2#
```

Figure 46

On the P–PE2 segment, the packet carries only the VPN label (value 19), as shown in Figure 43. This results from Penultimate Hop Popping (PHP): the P router removes the outer transport label before forwarding the packet to PE2. PHP is implemented through LDP by advertising the implicit-null label for directly connected prefixes. PE2’s LDP bindings confirm

this behavior, as the local binding for 2.2.2.2/32 is **imp-null**, instructing the P router to pop the transport label before delivery. Consequently, PE2 receives the packet with only the inner VPN label, avoiding an additional label lookup.

PE2 uses the VPN label to identify the destination VRF and forward the packet to the correct CE. This is confirmed by PE2's MPLS forwarding table in Figure 46, where label 19 maps to 192.168.102.0/24[V]. The [V] annotation indicates a VRF route, and the outgoing label field shows **No Label**, meaning PE2 removes the VPN label and forwards the packet natively into the blue VRF toward 10.10.2.100. Similarly, the outer transport label value 17 observed on the PE1-P link was assigned by the P router and advertised to PE1 through LDP.

The VPN label ensures isolation between organizations, even when overlapping IP address spaces are used. Since forwarding at the egress PE is based on the VPN label rather than the destination IP address, packets with identical IP destinations can be delivered to different VRFs without ambiguity. The outer transport label is swapped at each LSR hop across the provider core, while the inner VPN label remains unchanged from ingress PE to egress PE, where it is finally removed.

6.5 BGP Session Establishment and VPNv4 Route Advertisement

To analyze the BGP Open and Update messages exchanged between PE1 and PE2, interface f0/0 of PE2 was shut down and re-enabled while a Wireshark capture was running on the PE1-P link, forcing a new BGP session to be established.

Figure 47 presents the BGP Open messages sent by PE1.

78	57.104623	c2:03:d9:13:00:00	CDP/VTP/DTP/PAgP/UD...	CDP	348	Device ID: P	Port ID: FastEthernet0/0
79	60.028098	1.1.1.1	2.2.2.2	TCP	62	28111 → 179 [SYN] Seq=0 Win=16384 Len=0 MSS=1436	
80	60.077974	2.2.2.2	1.1.1.1	TCP	58	179 → 28111 [SYN, ACK] Seq=0 Ack=1 Win=16384 Len=0 MSS=1436	
81	60.098572	1.1.1.1	2.2.2.2	TCP	60	28111 → 179 [ACK] Seq=1 Ack=1 Win=16384 Len=0	
82	60.098606	1.1.1.1	2.2.2.2	BGP	119	OPEN Message	
83	60.128394	2.2.2.2	1.1.1.1	TCP	54	179 → 28111 [ACK] Seq=1 Ack=62 Win=16323 Len=0	
84	60.138509	2.2.2.2	1.1.1.1	BGP	115	OPEN Message	
85	60.149160	2.2.2.2	1.1.1.1	BGP	73	KEEPALIVE Message	
86	60.149486	1.1.1.1	2.2.2.2	BGP	77	KEEPALIVE Message	
87	60.351008	1.1.1.1	2.2.2.2	TCP	60	28111 → 179 [ACK] Seq=81 Ack=81 Win=16304 Len=0	
88	60.370897	2.2.2.2	1.1.1.1	TCP	54	179 → 28111 [ACK] Seq=81 Ack=81 Win=16304 Len=0	
89	60.481981	ca:01:d8:70:00:00	ca:01:d8:70:00:00	LOOP	60	Reply	
90	61.035360	00:00:00:00:00:00	00:00:00:00:00:00	CDP	64	Hello Packet	


```

MultiProtocol Label Switching Header, Label: 17, Exp: 6, S: 1, TTL: 255
Internet Protocol Version 4, Src: 1.1.1.1, Dst: 2.2.2.2
Transmission Control Protocol, Src Port: 28111, Dst Port: 179, Seq: 1, Ack: 1, Len: 61
Border Gateway Protocol - OPEN Message
  Marker: ffffffffffffffffffffffffffffffff
  Length: 61
  Type: OPEN Message (1)
  Version: 4
  My AS: 100
  Hold Time: 180
  BGP Identifier: 1.1.1.1
  Optional Parameters Length: 32
  Optional Parameters
    Optional Parameter: Capability
      Parameter Type: Capability (2)
      Parameter Length: 6
      Capability: Multiprotocol extensions capability
    Optional Parameter: Capability
      Parameter Type: Capability (2)
      Parameter Length: 6
      Capability: Multiprotocol extensions capability
    Optional Parameter: Capability
      Parameter Type: Capability (2)
      Parameter Length: 2
      Capability: Route refresh capability (Cisco)
    Optional Parameter: Capability
      Parameter Type: Capability (2)
      Parameter Length: 2
      Capability: Route refresh capability
    Optional Parameter: Capability
      Parameter Type: Capability (2)
      Parameter Length: 6
      Capability: Support for 4-octet AS number capability

```

Figure 47: Open messages exchanged

```

▶ Frame 88: 119 bytes on wire (952 bits), 119 bytes captured (952 bits) on interface -, id 0
▶ Ethernet II, Src: ca:01:d8:70:00:00 (ca:01:d8:70:00:00), Dst: c2:03:d9:13:00:00 (c2:03:d9:13:00:00)
▶ MultiProtocol Label Switching Header, Label: 16, Exp: 6, S: 1, TTL: 255
▶ Internet Protocol Version 4, Src: 1.1.1.1, Dst: 2.2.2.2
▶ Transmission Control Protocol, Src Port: 41407, Dst Port: 179, Seq: 1, Ack: 1, Len: 61
▼ Border Gateway Protocol - OPEN Message
  Marker: ffffffffffffffffffffffffffffffffff
  Length: 61
  Type: OPEN Message (1)
  Version: 4
  My AS: 100
  Hold Time: 180
  BGP Identifier: 1.1.1.1
  Optional Parameters Length: 32
  ▼ Optional Parameters
    ▼ Optional Parameter: Capability
      Parameter Type: Capability (2)
      Parameter Length: 6
      ▼ Capability: Multiprotocol extensions capability
        Type: Multiprotocol extensions capability (1)
        Length: 4
        AFI: IPv4 (1)
        Reserved: 00
        SAFI: Labeled VPN Unicast (128)
      ▼ Optional Parameter: Capability
        Parameter Type: Capability (2)
        Parameter Length: 6
        ▼ Capability: Multiprotocol extensions capability
          Type: Multiprotocol extensions capability (1)
          Length: 4
          AFI: IPv4 (1)
          Reserved: 00
          SAFI: Unicast (1)
      ▼ Optional Parameter: Capability
        Parameter Type: Capability (2)
        Parameter Length: 2
        ▼ Capability: Route refresh capability (Cisco)
          Type: Route refresh capability (Cisco) (128)
          Length: 0
      ▼ Optional Parameter: Capability
        Parameter Type: Capability (2)
        Parameter Length: 2
        ▼ Capability: Route refresh capability
          Type: Route refresh capability (2)
          Length: 0
      ▼ Optional Parameter: Capability

```

Figure 48: Content of the open message

The Open messages advertise the Multiprotocol Extensions capability (code 1) with AFI=1 (IPv4) and SAFI=128 (Labeled VPN Unicast). This is the critical capability that enables VPNv4 route exchange, standard BGP only supports IPv4 unicast (SAFI=1) and cannot carry prefixes with an attached Route Distinguisher or MPLS label. Without this capability being confirmed by both peers, no VPN routes can be signaled.

Figure 49 presents the BGP Update message sent by PE2 to PE1 advertising the blue organization's remote prefix. Two path attributes are of particular relevance, as shown in Figure 50. The MP_REACH_NLRI attribute (type 14) confirms the negotiated address family (AFI=1, SAFI=128) and carries the NLRI containing three elements: the Route Distinguisher 1:1 prepended to prefix 192.168.102.0, forming the 96-bit VPNv4 address, and VPN label 19 at the bottom of the stack. This label value directly matches the inner label observed in the previous section's Wireshark captures and PE2's MPLS forwarding table entry for 192.168.102.0/24[V], confirming that VPN labels are distributed via MP-BGP and not LDP. The next hop is encoded as RD=0:0, IPv4=2.2.2.2, identifying PE2's loopback as the

reachability next hop.

140	110.404141	2.2.2.2	1.1.1.1	TCP	54 179 → 28111 [ACK] Seq=81 Ack=100 Win=16285 Len=0
141	110.483714	ca:01:d8:70:00:00	ca:01:d8:70:00:00	LOOP	60 Reply
142	110.635689	2.2.2.2	1.1.1.1	BGP	73 KEEPALIVE Message
143	110.645842	2.2.2.2	1.1.1.1	BGP	234 UPDATE Message, UPDATE Message
144	110.655989	2.2.2.2	1.1.1.1	BGP	77 UPDATE Message
145	110.659061	1.1.1.1	2.2.2.2	TCP	60 28111 → 179 [ACK] Seq=100 Ack=280 Win=16105 Len=0
146	110.862103	1.1.1.1	2.2.2.2	TCP	60 28111 → 179 [ACK] Seq=100 Ack=303 Win=16082 Len=0
147	111.214433	1.1.1.1	2.2.2.2	BGP	238 UPDATE Message, UPDATE Message
148	111.214481	1.1.1.1	2.2.2.2	BGP	81 UPDATE Message
149	111.249857	2.2.2.2	1.1.1.1	TCP	54 179 → 28111 [ACK] Seq=303 Ack=303 Win=16082 Len=0
150	112.054094	222.222.1.10	224.0.0.2	LDP	76 Hello Message
151	113.709396	c2:03:d9:13:00:00	c2:03:d9:13:00:00	LOOP	60 Reply
152	113.709396	c2:03:d9:13:00:00	c2:03:d9:13:00:00	LOOP	76 Hello Message
Frame 143: 234 bytes on wire (1872 bits), 234 bytes captured (1872 bits) on interface -, id 0 Ethernet II, Src: c2:03:d9:13:00:00 (c2:03:d9:13:00:00), Dst: ca:01:d8:70:00:00 (ca:01:d8:70:00:00) Internet Protocol Version 4, Src: 2.2.2.2, Dst: 1.1.1.1 Transmission Control Protocol, Src Port: 179, Dst Port: 28111, Seq: 100, Ack: 100, Len: 180 Border Gateway Protocol - UPDATE Message Marker: ffffffffffffffffffffffffffffffff Length: 90 Type: UPDATE Message (2) Withdrawn Routes Length: 0 Total Path Attribute Length: 67 Path attributes Path Attribute - ORIGIN: INCOMPLETE Path Attribute - AS_PATH: empty Path Attribute - MULTI_EXIT_DISC: 0 Path Attribute - LOCAL_PREF: 100 Path Attribute - EXTENDED_COMMUNITIES Path Attribute - MP_REACH_NLRI Border Gateway Protocol - UPDATE Message Marker: ffffffffffffffffffffffffffffffff Length: 90 Type: UPDATE Message (2) Withdrawn Routes Length: 0 Total Path Attribute Length: 67 Path attributes Path Attribute - ORIGIN: INCOMPLETE Path Attribute - AS_PATH: empty Path Attribute - MULTI_EXIT_DISC: 0 Path Attribute - LOCAL_PREF: 100 Path Attribute - EXTENDED_COMMUNITIES Path Attribute - MP_REACH_NLRI					

Figure 49: Messages exchanged

```

> Frame 143: 234 bytes on wire (1872 bits), 234 bytes captured (1872 bits) on interface -, id 0
> Ethernet II, Src: c2:03:d9:13:00:00 (c2:03:d9:13:00:00), Dst: ca:01:d8:70:00:00 (ca:01:d8:70:00:00)
> Internet Protocol Version 4, Src: 2.2.2.2, Dst: 1.1.1.1
> Transmission Control Protocol, Src Port: 179, Dst Port: 28111, Seq: 100, Ack: 100, Len: 180
< Border Gateway Protocol - UPDATE Message
  Marker: ffffffffffffffffffffffffffffffff
  Length: 90
  Type: UPDATE Message (2)
  Withdrawn Routes Length: 0
  Total Path Attribute Length: 67
  < Path attributes
    > Path Attribute - ORIGIN: INCOMPLETE
    > Path Attribute - AS_PATH: empty
    > Path Attribute - MULTI_EXIT_DISC: 0
    > Path Attribute - LOCAL_PREF: 100
    < Path Attribute - EXTENDED_COMMUNITIES
      > Flags: 0xc0, Optional, Transitive, Complete
      Type Code: EXTENDED_COMMUNITIES (16)
      Length: 8
      < Carried extended communities: (1 community)
        < Route Target: 1:1 [Transitive 2-Octet AS-Specific]
          > Type: Transitive 2-Octet AS-Specific (0x00)
          Subtype (AS2): Route Target (0x02)
          2-Octet AS: 1
          4-Octet AN: 1
        < Path Attribute - MP_REACH_NLRI
          > Flags: 0x80, Optional, Non-transitive, Complete
          Type Code: MP_REACH_NLRI (14)
          Length: 32
          Address family identifier (AFI): IPv4 (1)
          Subsequent address family identifier (SAFI): Labeled VPN Unicast (128)
        < Next hop: RD=0:0 IPv4=2.2.2.2
          Route Distinguisher: 0:0
          IPv4 Address: 2.2.2.2
          Number of Subnetwork points of attachment (SNPA): 0
        < Network Layer Reachability Information (NLRI)
          < BGP Prefix
            Prefix Length: 112
            Label Stack: 19 (bottom)
            Route Distinguisher: 1:1
            MP Reach NLRI IPv4 prefix: 192.168.102.0

```

Figure 50: Contents of the message

The `EXTENDED_COMMUNITIES` attribute (type 16) carries Route Target `RT:1:1`. Upon receiving this update, PE1 imports the prefix into the blue VRF because its import policy matches `RT:1:1`, while the red VRF, configured with `route-target import 2:2`, ignores it entirely. This enforces inter-organization isolation at the control plane, complementing the data plane isolation provided by VRFs.

A VPNv4 route is an IPv4 prefix extended with a 64-bit Route Distinguisher, producing a globally unique 96-bit BGP prefix. The RD ensures overlapping customer address spaces remain distinguishable — `1:1:192.168.102.0/24` and `2:2:192.168.102.0/24` are distinct BGP entries despite identical IPv4 portions. Crucially, the RD carries no policy semantics; routing policy is determined solely by the Route Target. Standard BGP cannot support this VPN solution because it simultaneously lacks the 64-bit RD prefix format, MPLS label encoding within the NLRI, and Extended Communities support for Route Target signaling.